Residential Planning Skills

Usage Guide and Complete Skill Files

2026-10-09

Review Guide

Purpose and completeness

This document brings together three residential planning skills for review: `kitchen-planning`, `kitchen-planning-fr`, and `living-room-planning`. It includes a usage and comparison guide followed by the complete current contents of all nine maintained files, including YAML metadata, operational references, checklists, worked examples, source catalogs and evidence limitations. Nothing in the skill files is shortened for this review edition.

Markdown is rendered as Word headings, lists, tables, code and clickable links. The source file path and SHA-256 fingerprint at each chapter identify the exact snapshot. The Word Navigation Pane and the linked contents below support review. The Markdown originals remain authoritative; this Word document does not install or activate a skill. Including the catalogs here does not change their optional loading policy during normal use.

Which skill to use

Skill	Primary use	Important boundary
kitchen-planning	Residential kitchen layout, preparation, cabinet contents, dishwasher unloading and installation-review preparation; Germany/EU defaults.	Confirm the actual jurisdiction. No single EU kitchen code or universal sink/hob legal distance is assumed.
kitchen-planning-fr	Residential kitchens in France, with electrical, ventilation, tenure, rental, accessibility and quotation context.	Self-contained: does not require the German-default skill. French technical context does not imply a national style or household stereotype.
living-room-planning	Seating, circulation, conversation/media, storage, lighting and changing activity modes; Germany/EU defaults.	Check occupied and deployed states. A closed furniture footprint does not prove that a recliner or sofa bed works.

For a French kitchen, select the French specialization rather than merging conflicting country defaults. For an open-plan kitchen/living room, use the relevant kitchen skill with the living-room workflow and one shared measured plan, constraint register and circulation model. Re-establish local requirements for the living area; the French kitchen skill does not make the living-room skill a verified French regulatory specification.

File roles and loading

Each package has the same structure under `.github/skills/<skill-name>/`:

```
SKILL.md
references/
    planning-reference.md
    source-catalog.md
```

- `SKILL.md`: discovery metadata, scope, intake, workflow, decision rules and deliverables.
- `references/planning-reference.md`: qualified dimensions, operational checks and examples; consult as directed by the main skill, especially before numerical or readiness claims.
- `references/source-catalog.md`: provenance, URLs, source limitations and research notes; optional during routine planning, available for verification, citation questions or updates.

Project-specific regulations, exact product manuals and professional checks remain necessary where applicable. Optional bibliography loading does not waive those checks. Source IDs belong to their own package; identical IDs in the general kitchen and living-room packages do not identify the same source.

Invocation and required inputs

In a compatible VS Code Copilot host, invoke `/kitchen-planning`, `/kitchen-planning-fr`, or `/living-room-planning`, or describe a task matching the skill's discovery description. Discovery depends on host configuration. Other AI applications need their own instruction/resource-loading integration.

Supply the location, measured room and openings, fixed services, retained items, exact products where known, household activities, access needs, inventory, budget and constraints. Photographs can provide context, but cannot establish hidden services, wall strength, exact dimensions or installation readiness. Ask for a concept with explicit assumptions if those facts are not yet available.

Shared planning method

1. Establish the household brief and measured site facts.
2. Classify regulatory, manufacturer, site and user-must constraints separately from recommendations and preferences.
3. Design activity zones and assign real contents before selecting furniture dimensions.
4. Compare feasible alternatives using occupied, moving and open/deployed states.
5. Record evidence, failures, unknowns, remedies and responsible reviewers.
6. Deliver a dimensioned proposal, contents schedule, trade-offs, cost scope and next actions.

Unknown is never pass. Recommendations must not silently become laws, and preferences cannot override applicable mandatory requirements. Preserve the planner's authoritative geometry, catalog, prices and accepted design; proposed changes remain reversible until accepted.

Skill-specific review focus

General kitchen: Review the sink/preparation/hob relationship, usable worktop, exact appliance compatibility and cabinet contents. Test unloading with the dishwasher door, racks, receiving drawer and user all represented. Measure walkable routes rather than lines through furniture.

French kitchen: Review the applicable NF C 15-100 edition and work scope, ordinary sockets versus appliance circuits, hood versus dwelling ventilation, collective ducts, permissions, correct rental branch, French accessibility applicability and an itemized installed EUR TTC quotation. Technical and legal evidence gaps remain explicit; summaries are not full standards. The skill does not import German DIN dimensions as French requirements.

Living room: Review activities before assuming a TV-led layout. Test actual seating comfort, occupied routes, full recliner/sofa-bed deployment, parking for displaced furniture, storage access, screen glare, lighting, anchoring, cables, heating/ventilation and safe exit access in each required mode.

Example review requests

- "Use /kitchen-planning to compare two layouts from this measured plan. Allocate dishes and cooking tools and test a complete dishwasher-unloading cycle."
- "Use /kitchen-planning-fr for this French apartment renovation. Separate verified facts from electrician, ventilation, syndic and rental-equipment questions."
- "Use /living-room-planning to arrange conversation, television and occasional sleeping. Show recliner and sofa-bed states and where the coffee table moves."
- "Review this open-plan room with the relevant kitchen skill and living-room skill. Check shared routes while cooking and while all everyday seats are occupied."

Deliverables and readiness

Expect a brief and assumptions, dimensioned layout or explicit wall-by-wall arrangement, furniture/cabinet contents schedule, alternatives, task/mode checks, constraint register, budget scope and professional-review actions.

Status	Meaning for review
concept	Provisional geometry, products or unresolved information remain visible.
dimensionally_checked	Only the stated geometry, products, modes and checks supported by evidence are covered; disclose remaining technical or regulatory unknowns.
ready_for_professional_order_review	Required evidence is assembled and relevant hard failures/unknowns are resolved, subject to the specific skill's gate. Professional review and purchasing authorization remain separate.

None of these labels certifies installation, safety, accessibility or legal compliance. These skills are written guidance, not deterministic geometry engines, product databases or substitutes for qualified professional work.

Reviewer checklist

- Confirm scope, jurisdiction and country-specific branching are correct.
- Check whether numerical statements identify their measurement definition and authority.
- Challenge false passes involving unknown services, open furniture, loaded storage or blocked access.
- Verify source applicability and currency for disputed or high-consequence claims.
- Confirm deliverables expose uncertainty rather than implying order authorization.
- Record feedback against the source path, heading and rule ID shown in the complete-file chapters.

Maintenance and reproduction

Make accepted changes in the Markdown originals and regenerate this snapshot:

```
python scripts/build-kitchen-documentation.py --all
```

The builder requires `python-docx`, `pypandoc` and Pandoc. The command without `--all` retains the original single-kitchen documentation mode. The combined build includes all expected files, validates source-text coverage, internal bookmarks and Word package XML, and verifies that source files did not change during generation. These checks do not replace a visual review in Word or professional review of the planning guidance.

Complete File Contents

The following chapters reproduce the maintained files in full. Repeated guidance is intentionally retained so each skill can be reviewed independently.

- kitchen-planning: Full Skill Definition
- kitchen-planning: Full Planning Reference
- kitchen-planning: Full Source Catalog
- kitchen-planning-fr: Full Skill Definition
- kitchen-planning-fr: Full Planning Reference
- kitchen-planning-fr: Full Source Catalog
- living-room-planning: Full Skill Definition
- living-room-planning: Full Planning Reference
- living-room-planning: Full Source Catalog

kitchen-planning: Full Skill Definition

Source file: `.github/skills/kitchen-planning/SKILL.md`

SHA-256: `f7625a8313a3654b86d65aad3f17c095b125798e453d238c157cab0181b73200`

```
---
name: kitchen-planning
description: 'Plan or review a residential kitchen: room survey, kitchen layout, sink and hob spacing, dishwashe
r placement, cabinet ordering, drawer organization, storage zones, short walking routes, ergonomics, accessibili
ty, appliance clearances, and installation readiness. Use for kitchen planning, kitchen design, Kuechenplanung,
arranging cabinets, or improving cooking and dishwasher-unloading workflows. Defaults to Germany/EU and millimet
res; adapt rules to the actual jurisdiction.'
---
```

Kitchen Planning

Purpose and Scope

Produce practical, dimensioned residential kitchen proposals with short daily routes, usable storage, and explicit feasibility checks. Plan from activities and contents before choosing cabinet fronts. Here, cabinet "ordering" means arranging cabinets; procurement requires a separate approval gate.

Default to Germany/EU, dimensions in mm, and costs in EUR. Confirm country and, in Germany, Bundesland before applying regulatory requirements. There is no single EU kitchen-clearance code. Commercial kitchens require a separate specialist workflow.

Read the planning constraints reference before applying numerical guidance or reviewing installation risks. It includes rule classifications, qualifications, and short source IDs for traceability. Recheck current local requirements and exact product manuals for each real project.

The source catalog is optional. Load it only to verify a source-backed claim, answer a provenance/citation question, or update the guidance. Do not load it during routine planning merely to resolve source IDs; the operational guidance and its limitations are in this skill and the planning reference. This does not waive project-specific verification of regulations or product instructions.

Non-Negotiable Planning Discipline

- Treat law, applicable technical requirements, product instructions, and verified site constraints as simultaneous constraints. If they conflict, stop and resolve the conflict; a manufacturer cannot waive the law.
- Do not turn a recommendation into a regulation. In particular, do not claim that a sink beside a hob is universally illegal. Usually the problem is missing preparation/landing space; actual installation restrictions are product- and site-specific.
- Unknown is not pass. Missing dimensions, manuals, or service information remain unresolved. Photographs and generated images cannot establish installation readiness.
- Use a planner's authoritative geometry, catalog, and prices where available. Never invent a catalog item, price, service point, clearance, or successful validation result.
- Optimize only after hard constraints pass. A short walking route cannot compensate for a blocked exit, unsafe installation, or inaccessible appliance.
- Offer concepts with explicit assumptions when facts are incomplete. Do not call them order-ready or installation-ready.
- Do not move services, alter structures, purchase items, or overwrite an accepted design without explicit authorization. Draft alternatives should remain reversible.

1. Collect the Brief and Survey

Ask only for missing information that changes the next decision. Start with room dimensions/openings, fixed services, users, appliances, and budget. Collect remaining details progressively.

Input	Required information
Location and scope	Country, region, dwelling type, new build/remodel, ownership/landlord restrictions, required accessibility standard, retained items.
Room	Each wall length, ceiling/soffit heights, angles/diagonals, niches, columns, slopes, and finished floor levels. Measure irregular walls at relevant heights.
Openings	Door/window position, clear opening, frame/trim, sill height, opening direction and swept volume; curtains/blinds and window handles.
Services	Measured water/drain positions and heights, shutoffs, electrical points/circuits, gas, ventilation/exhaust routes, heating, meters, and service access. Mark relocation permission separately.
Structure	Wall construction/fixing capacity, floor/ceiling constraints, existing ventilation, room-air-dependent fireplace or combustion appliance, underfloor services.
People	Household size, number of simultaneous cooks, heights/elbow heights, handedness, reach/lifting ability, mobility aids, children, and future needs.
Activities	Shopping frequency, everyday meals, batch cooking, baking, coffee, entertaining, table setting, cleaning, waste sorting, and current annoyances.
Inventory	Counts and dimensions of dishes, pans, trays, containers, food, appliances, and waste bins; frequency and typical weight of use.
Products	Exact models or provisional envelopes, installation manuals, cabinet system, appliance niches, front/hinge system, handle projections, and worktop material.
Commercial	Total installed budget, service relocation allowance, delivery/access constraints, timing, and priorities ranked by the user.

For each critical fact record `value`, `unit`, `source`, `verification_state`, and measurement uncertainty where known. Use `unknown`, `inferred`, `user_verified`, or `professionally_verified`. Preserve raw measurements and identify the finished reference surfaces.

If only a photo is supplied, request a measured plan or verified reference dimensions. Hidden pipe routes, wall strength, electrical capacity, and regulatory compliance cannot be inferred from appearance.

2. Build the Constraint Register

Give each rule a stable ID and classify it:

Class	Meaning	Handling
`regulatory`	Verified applicable legal/code requirement, with jurisdiction and edition.	Hard; no user override.
`manufacturer`	Exact installation/compatibility requirement for the selected product.	Hard; change the product or design if necessary.
`site`	Verified room, structural, service, or geometric limitation.	Hard unless an approved change creates a new verified condition.
`user_must`	Explicit user requirement, including a fixed budget if stated.	Hard for this brief; ask before relaxing.

Class	Meaning	Handling
`recommendation`	Sourced ergonomic/design recommendation or labeled planning heuristic.	Soft; document the consequence of departures.
`preference`	Style, symmetry, convenience, or other discretionary choice.	Soft; trade off transparently.

Each record needs `id`, `class`, `applicability`, `requirement`, `measurement_definition`, `source`, `status`, `affected_items`, and `remedy_or_next_action`. Status is `pass`, `fail`, `unknown`, or `not_applicable`; explain exclusions. Preserve recommendation deviations rather than silently deleting them.

Complete every category in the reference's installation checklist. Record applicability separately from compliance: an unconfirmed rule is not automatically irrelevant.

3. Arrange Functional Zones

Use five zones as an adjacency model, not a compulsory left-to-right sequence [S1, S2]:

1. Food storage: refrigerator/freezer and dry provisions.
2. Preparation: the principal clear work surface and preparation tools.
3. Cooking: hob, oven, heat-safe landing surfaces, and cookware.
4. Cleaning: sink, dishwasher, waste sorting, and cleaning supplies.
5. Clean storage/serving: dishes, dining cutlery, glasses, and serving items.

Trace these distinct workflows:

- Shopping arrival -> grocery landing -> refrigerator/pantry.
- Food storage -> sink when needed -> preparation -> cooking -> serving.
- Table return -> scrape/sort -> sink as needed -> dishwasher. Do not prescribe routine pre-rinsing; follow appliance guidance.
- Dishwasher -> clean storage -> table/serving.
- Hob -> sink for draining hot cookware, without avoidable traffic crossings.
- Coffee/breakfast and snack access without walking through active cooking.

Keep through-traffic outside the principal work area. The refrigerator-sink-hob triangle is a diagnostic, not a mandatory geometry or a substitute for these workflows. A one-wall kitchen can work without a triangle. Mixed-handed or multi-cook households need task trials, not automatic mirroring.

Sink, Preparation Surface, and Hob

Prefer `sink -> continuous preparation surface -> hob`, or the mirror image. Blum recommends approximately 900 mm of preparation width, with 800 mm as its lower ergonomic recommendation [S3]. These are not German legal minimums. More space may be needed for two cooks or extensive baking.

Measure usable clear surface between fixture edges, excluding an occupied drainer, permanent appliances, or other obstructions. Record actual usable depth too. Separately verify cut-out-to-cut-out webs, edge distances, and supports from the worktop fabricator; a visible surface gap does not prove structural suitability.

Avoid placing a tall cabinet across this continuous workflow. Keep the preparation surface genuinely available after locating the kettle, coffee machine, knife storage, and drying area. Where the room cannot accommodate the recommendation, disclose the compromise and test an adjacent alternative preparation surface.

Dishwasher and Clean Storage

- Start with the dishwasher beside the sink, subject to plumbing, appliance instructions, and usable access. Prefer a side outside the main preparation stance when feasible.
- Compare both sides of the sink. Do not prescribe left/right solely from handedness.
- Locate everyday dining cutlery, plates, and bowls in an adjacent or nearby accessible drawer bank. Locate daily glasses and cups within comfortable reach nearby [S4].
- Aim for a comfortable turn or short step during unloading. This is a qualitative target, not a universal distance limit.
- Check dishwasher fully open, racks extended, destination drawer open, and a person standing in the unloading position. Neither nominal adjacency nor cabinet-centre distance establishes usability.
- Across an aisle can work only if the actual door/drawer projections, standing space, reach, and circulation work together. Never require stepping over a dishwasher door.
- At an inside corner, check perpendicular drawer collisions and access to the sink/waste. Move the dishwasher or storage bank if fillers alone cannot solve the conflict.
- Keep dining/table access convenient without routing other people through the hot zone. Do not put every item beside the dishwasher: cooking tools still belong at their principal point of use.
- Consider a raised dishwasher for users who benefit, but validate approved support, niche, front mechanism, rack heights, and adjacent unloading space.

4. Assign Contents Before Cabinet Sizes

Create the storage schedule first. Then fit catalog modules around the fixed zones.

Contents	Preferred home	Checks
Dining cutlery	Shallow upper drawer near dishwasher and dish storage.	Organizer fits internally; accessible with dishwasher open.
Everyday plates/bowls	Full-extension drawer in the same or nearby bank.	Plate diameter, stack height, restraint, loaded runner capacity, comfortable lifting height.
Glasses/cups	Reachable nearby shelf, lift-front cabinet, or suitable drawer.	Safe reach, glass height, retention, opening clearance; not forced overhead.
Preparation knives/tools	Upper drawer at principal preparation surface.	Blade protection, child security, separation from dining cutlery.
Boards/trays	Vertical divider storage near preparation/oven.	Actual tray dimensions and removal path.
Pans/pots/lids	Full-extension storage beside or below hob where permitted.	Heat separation, hob underside, ventilation and extractor/oven volume.
Spices/oils	Near preparation/cooking, protected from heat/light/moisture.	Not directly exposed to hob heat or steam.
Mixing bowls, scales, baking tools	Preparation/baking zone.	Nesting, appliance weight, lifting distance, and usable drawer height.
Dry food	Pantry/pull-outs near grocery landing and preparation.	Stock volume, visibility, shelf adjustment, runner loads.
Waste/recycling	Accessible from preparation and sink.	Local sorting streams, bin removal, odours, and access while dishwasher is open.
Cleaning chemicals	Separate secure storage, often near sink.	No food/clean-utensil mixing; child access, plumbing and leak clearance.
Food containers/lids	Near preparation, leftovers packing, and dishwasher.	Paired or divided organization; capacity for actual inventory.

Contents	Preferred home	Checks
Small appliances	At use location or reachable storage.	Operating footprint, lid opening, steam, power, ventilation, weight.
Rarely used items	Less accessible but safe locations.	Do not put heavy objects overhead simply because they are infrequent.

Prefer full-extension drawers for frequently used base storage [S4, S5]. Upper shallow drawers suit small tools; deeper drawers suit bulky items, but choose heights from actual contents, not a fixed three-drawer formula. Prioritize heavy frequently used items at a comfortable lifting height rather than automatically on the lowest level.

Compare wide drawer banks with multiple narrow cabinets using usable capacity, load, cost, and multi-user access. A wide open drawer can block another cook. Compare corner mechanisms with a deliberately unused corner plus accessible straight drawers; maximum gross volume is not necessarily best value.

For each cabinet record: `id`, `zone`, `catalog_id_or_provisional`, external dimensions, usable internal dimensions, contents, load, opening direction/extension, fittings, and reason for location. Include fillers, end panels, scribes, plinths, supports, and tolerances in each run. Never assume nominal width equals usable storage or appliance niche width.

5. Generate and Check Alternatives

Develop two or three genuinely different feasible options when geometry permits, for example retained services versus improved workflow. Do not force an island or a U-layout into a room that cannot support it.

For each option:

1. Place immovable constraints and reserve circulation/accessibility envelopes.
2. Locate sink, principal preparation area, hob/extraction, and refrigerator landing.
3. Fit dishwasher, waste, and nearby clean storage as a working group.
4. Allocate remaining storage and secondary tasks from the inventory.
5. Fit actual modules, fillers, worktops, panels, and service voids.
6. Check 3D fit, closed projections, full swept openings, service access, ventilation, and installation tolerances.
7. Run the task trials below. Repair hard failures before comparing soft preferences.

Required trials: unload a full dishwasher; prepare/cook a typical meal; drain a heavy pot; set/clear the table; put away groceries; remove waste; and let a second person reach drinks while cooking continues. Include the actual accessibility/child-use scenarios in the brief.

Check necessary simultaneous openings, especially dishwasher plus receiving drawer, refrigerator door plus internal drawer, sink/waste plus dishwasher, and perpendicular corner units. Not every door must open simultaneously, but the normal task combinations must work. Record remaining conflicts and their consequences.

Measure travel along collision-free walkable paths between usable standing positions, with task-specific openings represented. Do not measure straight through an island or across an open door. Without a geometry/path tool, use a clearly labeled manual estimate or `unknown`; do not fabricate precision.

Compare Everyday Effort

Use lexicographic priorities: hard feasibility first, essential usability/storage second, daily effort third, then cost/style trade-offs within the agreed brief.

For a transparent optional distance estimate, calculate:

```
daily_travel_m = sum(task_cycles_per_day * sum(trip_count_per_leg * walkable_leg_length_m))
```

Define each leg as a one-way movement. Include return legs explicitly where they occur. For unloading, use realistic batches carried per trip, not one trip per plate and not an implausibly heavy stack. Use the same inventory, carry capacity, and task frequencies for every option. Label assumed frequencies and distances.

Report bending/high reaching, awkward lifts, door conflicts, and traffic crossings separately; do not add metres, counts, and money without explicit normalization and user-agreed weights. Never label a heuristic score a certified optimum or claim percentage savings without a baseline calculation.

6. Deliver the Plan

Return a concise recommendation followed by:

1. Brief, assumptions, missing measurements, and readiness status.
2. Dimensioned layout, orientation, appliance locations, and clear work surfaces. If no drawing tool exists, supply wall-by-wall cabinet sequences with dimensions, not an invented rendered plan.
3. Cabinet/drawer schedule specifying exactly what goes where.
4. Dishwasher-unloading explanation: standing position, receiving drawers/cup storage, openings, paths, and compromises.
5. Alternatives table: hard failures/unknowns, preparation space, usable storage, travel estimate, ergonomics, installed cost confidence, and trade-offs.
6. Constraint report with rule IDs, measurement definitions, evidence, pass/fail/unknown, and remedies.
7. Action list assigning survey, electrical, plumbing, ventilation, structural, accessibility, and installation checks to the appropriate reviewer.

Use the status `concept`, `dimensionally_checked`, or `ready_for_professional_order_review`. The last status requires verified survey, exact product compatibility, complete parts/cost scope, and no unresolved hard constraint. It is not professional installation approval. When required reviewers approve, record who approved what and against which plan/manual revisions before any separate order authorization.

If no feasible layout exists, identify the conflicting constraints and smallest changes to consider, such as reducing an island, changing an appliance, or obtaining approval to relocate services. Never silently shrink a required clearance or alter a user must-have.

Quality Checks for This Skill

Before delivering a plan, verify these counterexamples are handled correctly:

- Sink beside hob: report preparation-space and installation checks, not an invented EU prohibition.
- Dish drawer close to dishwasher but behind its open door: reject the unloading arrangement despite the short nominal distance.
- Short route through an island: replace it with a walkable route or mark distance unknown.
- Attractive layout with unknown hood clearance: keep installation readiness blocked.
- Single-wall or mixed-handed kitchen: use task-based zoning without forcing a triangle or handedness rule.
- Wheelchair brief: invoke applicable accessibility checks and user trials; a generic aisle recommendation is insufficient.
- Best geometric score but insufficient plate storage or overloaded drawers: do not recommend until capacity/load are resolved.

kitchen-planning: Full Planning Reference

Source file: `.github/skills/kitchen-planning/references/planning-reference.md`

SHA-256: `b0c2f88236f9af433ade37b2317d6952a45b31b4ef7d79dfb1fae5e940658c85`

Kitchen Planning Evidence and Constraints

Research checked: 2026-10-09. Scope: residential kitchens, Germany/EU default. This is an original synthesis of public guidance, not a reproduction of a standard, a legal opinion, or an installation specification.

Evidence Boundaries

AMK, Blum, and IKEA offer useful planning guidance; their editorial advice is not law. Product marketing is not proof of compatibility or universal ergonomic superiority. A named standard is mandatory only where applicable through law, contract, or the project brief; establish that applicability with qualified reviewers.

Short source IDs refer to the optional source catalog, which holds URLs, publication details, and research notes. Do not load it for routine planning; consult it only for source verification, provenance/citation questions, or guidance updates. Rule classifications and operational qualifications remain in this reference.

The full DIN standard and project-specific appliance installation manuals were not examined. Numerical clearance requirements for a real product therefore remain to be supplied. Verify current applicable requirements and exact product instructions for each project; the research date does not establish a document's technical currency.

Some general advice is too absolute to reuse literally. For example, induction surfaces can remain hot from cookware, lift-up doors still need head/ceiling/reach checks, and handedness does not determine a whole household's best layout. Apply task and product checks rather than repeating marketing assurances.

Dimensions and How to Use Them

All dimensions below are in mm. Label the rule class and source whenever presenting them to a customer.

Topic	Starting guidance	Status and measurement
Main preparation width	Approximately 900; Blum advises at least 800 for comfortable preparation.	Published ergonomic recommendation [S3], not a statutory minimum. Clear usable work surface, preferably between sink and hob; not centre-to-centre distance.
Single-cook working aisle	Explore roughly 1000-1200 initially.	This skill's concept-sizing heuristic, not a sourced legal minimum. Measure the narrowest gap between finished closed projections, then test open appliances and the user.
Shared working aisle	Explore roughly 1200-1500 initially.	Concept-sizing heuristic only. Simultaneous work, carry tasks, and passage can need more; extra width also lengthens transfers.
Standing preparation height	Start near measured elbow height minus 150.	AMK recommendation [S1]; test in usual footwear with a mock-up. Sink depth and cookware change effective task height. No universal 900-high worktop.
Accessible kitchen movement space	The DIN 18040-2 secondary summary describes 1200 x 1200, and 1500 x 1500 for wheelchair use; front clearance depths of 1200 and 1500 respectively.	[S7] is a secondary explanation, not the standard itself. Confirm current applicable provisions, including the wheelchair-specific category, and all other access requirements. These are not universal minimum kitchen dimensions.

Topic	Starting guidance	Status and measurement
Dishwasher to dish storage	Comfortable turn or short step where feasible.	Adjacency principle [S4]; no universal maximum distance established by this research. Open-state usability determines success.
Hob/oven/fridge landing	Reserve usable nearby surface sized for the actual pan, tray, or grocery load.	Functional planning recommendation. Do not invent a universal legal landing width. The hot-tray route must be safe with the appliance door open.
Hood, hob, combustible surfaces	Use exact product and applicable local requirements.	No generic numeric default. Check all applicable instructions; resolving a conflict may require changing products.
Cut-out webs, overhangs, supports	Use material/fabricator specifications.	No generic numeric default; material, thickness, cut-outs, support and installation details matter.
Drawers, fronts, fillers, service voids	Use selected cabinet/appliance system drawings.	Catalog-specific; never insert an arbitrary filler and assume collision/installation issues are resolved.

A concept aisle width is not the free width remaining during use. For an illustrative opposing arrangement only, a 1200 aisle minus a 600 dishwasher-door projection minus a 500 drawer projection leaves just 100 between the open fronts. That is not adequate standing space between them. Real doors, racks, handles, body envelopes, and sideways access require actual drawings and task simulation. The numbers here are hypothetical, not product specifications.

Installation and Usability Checklist

Use the IDs below in the project's constraint register. These are check categories: obtain evidence to establish the exact hard requirement or recommendation for each project. Mark unknowns explicitly.

ID	Check	Evidence or required outcome
GEO-01	Verified geometry	Finished dimensions, out-of-square walls, slopes, trims, levels, and measurement tolerance are represented. No fit claim based only on photos.
GEO-02	Circulation and exit	Doors, clear passages, room access and applicable escape requirements remain usable. Assess appliance-open task states and occupants.
GEO-03	Collision volumes	Check full door/drawer travel, handles, lift fronts, internal pull-outs, adjacent fronts, and head clearance. Closed-box collision tests are insufficient.
GEO-04	Windows and heat	Tap does not prevent required window operation. Window/blind/curtain position, hob heat, drafts, splash exposure and required access are reviewed.
GEO-05	Seating and island	Account for occupied seats, pulled-out chairs, knee space, passage and hot/splash zones. Do not double-count the same area for incompatible simultaneous uses.
FIT-01	Cabinet run and niches	Include end panels, corner solutions, fillers, scribes, wall tolerances and actual appliance niches, not just nominal cabinet widths.
FIT-02	Front and plinth systems	Integrated dishwasher front weight/height, fixed or sliding hinge, plinth sweep, mounting and countertop protection match the exact model.
FIT-03	Refrigerator operation	Required opening angle permits internal drawers/shelves to operate and be removed; wall stand-off, hinges and ventilation are correct.
FIT-04	Hob underside and extraction	Body, induction cooling, required separation, cables, downdraft motor/duct/filter and drawers do not compete for the same volume.
FIT-05	Oven and microwave	Approved niche, ventilation, door travel and shelf reach; hot vessels can be removed safely to a landing surface. A high oven must not require lifting hot trays above the user's safe reach.
FIT-06	Raised appliances	Support structure, fixing, manufacturer approval and reachable controls/racks are verified. Do not put a floor-supported appliance on an improvised shelf.

ID	Check	Evidence or required outcome
WAT-01	Sink assembly	Bowl, drainer, clips, tap, waste, trap, hoses, drawers and bins fit together; servicing remains possible.
WAT-02	Water and drainage	Qualified installer verifies supplies, pressure, hot-water compatibility, drainage route/levels, appliance hose limits and required backflow protection. Proximity alone proves none of these.
WAT-03	Shutoff and leak access	Isolation, connections and leak-protection equipment remain accessible as required; no pinched hoses or inaccessible maintenance points.
WAT-04	Special water equipment	Boiling/chilled/carbonated/filter systems have approved taps, tank/filter/cylinder space, ventilation, power, maintenance and child protection.
ELE-01	Electrical design	Qualified electrician verifies available supply, connected loads, circuits, protective devices, earthing, isolation and socket installation under applicable rules.
ELE-02	Socket and cable placement	Plan for all fixed and small appliances, lighting and powered fittings. Keep cords out of wet/hot areas and across no circulation route; verify plugs do not consume a prohibited niche/service volume.
ELE-03	No invented wet-area rule	Do not transpose bathroom electrical zones into a kitchen or assert a blanket socket-to-sink distance without an applicable source. Obtain the electrician's project-specific layout.
GAS-01	Gas equipment	Approved professional checks supply, isolation, installation, ventilation and appliance-specific combustible clearances. No DIY connection instructions.
VEN-01	Extraction choice	Establish exhaust or recirculation early. Check hood capture at the actual hob, installed airflow, noise, filter access, duct dimensions, bends, outlet and replacement-air strategy.
VEN-02	Recirculation limits	Recirculation normally does not remove cooking moisture outdoors and is not a substitute for necessary room ventilation. Check system-specific requirements.
VEN-03	Combustion interaction	Exhaust plus a room-air-dependent fireplace/heater triggers professional pressure/backdraft review with the responsible chimney sweep and ventilation specialist. Do not assume a window switch alone makes the system safe.
VEN-04	Building permissions	Check landlord/owners' association approvals, facade penetrations, fire separations, communal shafts and whole-dwelling ventilation interactions.
TOP-01	Worktop structure	Fabricator verifies cut-out spacing, edge webs, joint locations, supports, spans, overhangs, thermal limits and installation method.
TOP-02	Water and steam protection	Seal/protect cut-outs, joints, sink area and dishwasher edges as specified; account for automatic-opening dishwasher steam.
STR-01	Fixings and stability	Suitable wall fixings, rails, suspended loads, tall-unit restraint, island support and loaded drawers. Check floor/underfloor services before drilling.
STR-02	Structural alterations	No wall removal or structural drilling assumption without qualified assessment and required permissions.
USE-01	Preparation and landings	Real unobstructed preparation and grocery/hot-vessel landing space remains after permanent countertop items are placed.
USE-02	Unloading	Dishwasher, rack, receiving storage and user can occupy needed positions together; no stepping over the door or excessive reaching.
USE-03	Waste and hygiene	Scrape/sort/load route works; bins can be removed and cleaned. Separate chemicals from food/clean items; provide cleanable food-contact surfaces and drying space.
USE-04	Storage capacity and loads	Actual inventory fits internally with chosen organizers and runner ratings. Record spare capacity if requested; do not invent a mandatory percentage.
USE-05	Multi-user tasks	Test concurrent cooking, dish handling, drinks access and table setting; identify crossings and bottlenecks.

ID	Check	Evidence or required outcome
ACC-01	Accessibility	Confirm required standard/category; check turning and approach spaces, knee/toe clearance, seated reach, controls, grip/force, task height and protected hot pipes. Test with the actual user/mobility aid.
SAFE-01	Children and burns	Secure knives/chemicals, review reachable controls and hot-water dispensing, prevent pot-handle exposure to traffic/seating. Induction still has residual heat; child locks do not replace supervision.
SAFE-02	Flooring and edges	Review slip/trip risk, thresholds, protrusions, sharp corners and cleanability. Do not let loose mats create an access hazard.
LGT-01	Lighting	Illuminate preparation, sink and hob without the user casting a working shadow. Check glare, contrast, controls and usable light beneath cabinets; lighting drivers remain serviceable.
ENV-01	Comfort and maintenance	Compare actual appliance noise, energy/water data, extractor filter costs, cleanability, material heat/moisture resistance and repairability. Verify current product labels.
SRV-01	Replacement access	Appliances can be isolated, removed and replaced after flooring/worktops are installed; filters, traps, drivers and fixings can be serviced.
ORD-01	Complete order scope	Include appliances, cabinets, fronts, hinges, runners, handles, fillers, panels, plinths, feet, rails, supports, worktops, cut-outs, lighting, extraction, fittings, delivery, installation and service work.
ORD-02	Release gate	Final survey, product/manual revisions, costs, permissions and responsible professional checks match the exact proposed order. Obtain explicit order authorization.

Worked Arrangement: Pattern, Not Template

An illustrative working run can be:

```
[dish/cutlery drawers] [dishwasher] [sink] [prep drawers + clear worktop] [hob/cookware]
```

Daily cups/glasses can occupy comfortably reachable storage above the dish bank or another nearby accessible location. Waste can be under the sink or in a separate nearby pull-out if loading, preparation and bin removal all work. Do not consume the only preparation drawer space without a replacement.

This pattern places clean storage on one side of cleanup and preparation on the other. It is neither a complete kitchen nor a mandatory sequence: refrigerator/pantry, oven, landing surfaces, end conditions and services must still be located. Mirror or split the run when the room and users benefit. A hob at the run's exposed end still needs its verified clearances and usable landing arrangement.

For an L-layout, a drawer around the corner from the dishwasher may be physically close but impossible to open during unloading. For a galley, opposing dish drawers may work or may block the user's standing position. The decisive check is the open-state task, not the diagram's apparent neatness.

Transparent Travel Example

Suppose two otherwise feasible options each need four storage transfers per dishwasher cycle. Assume three return legs to the dishwasher before the final transfer, one cycle daily, and equal one-way path lengths in either direction:

- Option A: 1.0 m per leg -> `(4 + 3) * 1.0 = 7.0 m` per cycle.
- Option B: 2.5 m per leg -> `(4 + 3) * 2.5 = 17.5 m` per cycle.
- Difference: 10.5 m per cycle under these assumptions, excluding any final return and other kitchen tasks.

These are illustrative distances and batch counts, not research results. Do not extrapolate time or lifetime savings without measurements. A comfortable turn may produce almost no walking but still involve awkward reaching; report that separately. An option with an obstructed unloading path is infeasible regardless of its calculated distance.

Maintenance and Unverified Requirements

The checklist is deliberately broad but not exhaustive. Actual regulatory requirements, permitted clearances, and compatibility depend on the site, products and users. Obtain the current applicable primary sources, exact installation manuals and qualified reviews before converting any unresolved checklist item into a pass.

Revisit this reference when products, material systems, regulations or source guidance change. Keep the general workflow vendor-neutral; replace examples with the project's authorized catalog without losing the open-door, capacity, accessibility and safety checks.

kitchen-planning: Full Source Catalog

Source file: `.github/skills/kitchen-planning/references/source-catalog.md`

SHA-256: `8a0fa7286401c34879f97d8e08e1fbef6b6c5c72d9742f578cf53e4475808a34`

Kitchen Planning Source Catalog

Optional research reference. Load only to verify a source-backed claim, answer a provenance/citation question, or update the guidance. Routine planning does not require this file; the rules, classifications, and operational qualifications are in the skill and planning reference.

Research checked: 2026-10-09. Scope: residential kitchens, Germany/EU default. The sources below were retrieved for the stated topics. The full DIN standard, paid NKBA handbook, and project-specific appliance installation manuals were not examined. The retrieval date does not establish technical currency; verify current applicable requirements and exact product instructions for each real project.

Sources

S1. AMK: Planen und gestalten

https://www.amk-ratgeber-kueche.de/kueche/planen-gestalten

German kitchen-industry association guidance. Supports functional zones, point-of-use storage, approximately 150 mm elbow-to-worktop difference, individualized working heights, full-extension drawers, early service planning, lighting and accessibility considerations. This is consumer guidance, not a regulation or a complete technical standard. Do not adopt every safety/marketing statement literally; see the evidence boundaries in the planning reference.

S2. IKEA Germany: Gestaltungsideen fuer reibungslose Ablaeufe

https://www.ikea.com/de/de/rooms/kitchen/how-to/kueche-effizient-planen-pub6d21e890/

Supports generous preparation space between sink and hob, proximity of waste and dishwasher, and placing cookware/tools at their point of use. Uses the work triangle as a general planning aid. Does not establish a statutory sink-to-hob distance.

S3. Blum: Speisenvorbereitung leicht gemacht

https://www.blum-inspirations.com/de-eu/wissen/speisenvorbereitung-leicht-gemacht-der-passende-schrank-unterstuetzt-sie-dabei

Manufacturer ergonomic guidance, article dated March 2022. Recommends a preparation area between sink and hob, approximately 900 mm wide, and advises at least 800 mm; preparation tools and bowls belong nearby in organized drawers. These dimensions are ergonomic recommendations, not proof of product clearance or German code compliance.

S4. Blum: Geschirr perfekt organisiert

https://www.blum-inspirations.com/de-eu/wissen/geschirr-perfekt-organisiert-im-unterschrank-mit-auszuegen

Manufacturer guidance, article dated June 2022. Supports dining cutlery and dishes together in full-extension base storage, nearby cups/glasses, and proximity to the dishwasher to reduce unloading travel. It does not supply a universal maximum distance or guarantee collision-free access.

S5. Blum: Die ergonomische Kueche

https://www.blum-inspirations.com/de-eu/wissen/die-ergonomische-kueche-das-muss-man-beachten

Manufacturer guidance, article dated July 2021. Explains access benefits of full-extension base drawers and overhead lift fronts versus projecting doors. Benefits still depend on actual reach, opening height, operating force and room geometry; powered fittings are optional, not necessary for every accessible kitchen.

S6. AMK: Geraetetechnik

https://www.amk-ratgeber-kueche.de/kueche/geraetetechnik

Supports early extraction planning, the importance of capture efficiency, storage consumed by downdraft equipment, and appliance feature/energy/noise considerations. Room-volume airflow rules of thumb alone do not constitute a ventilation design. The combustion-safety checklist in the planning reference is a professional-review trigger, not a quoted clearance rule from this page.

S7. nullbarriere.de: DIN 18040-2 Kuechen, Essplatz

https://nullbarriere.de/din18040-2-kuechen.htm

Specialist secondary summary of German dwelling accessibility guidance. Describes 1200/1500 movement-space dimensions and wheelchair considerations. It is not the official DIN text. Verify the current standard, project category, legal/contractual applicability and complete requirements before claiming conformity.

S8. IKEA Germany: Kuechenplanung

https://www.ikea.com/de/de/rooms/kitchen/kuechenplanung-pub6003162f/

Supports surveying in mm, marking openings/services, and staged planning, checking, ordering and installation. Links to current system-specific planning and installation documents. Those documents and the chosen product manuals must be obtained for a real order.

S9. IKEA Germany: Schubladen organisieren

https://www.ikea.com/de/de/rooms/kitchen/how-to/kuechenschraenke-und-schubladen-organisieren-8-ideen-die-zeit-und-platz-sparen-pube8ed7a50/

Supports full-extension storage, functional grouping, protected knives, organizers and reachable everyday contents. Named fittings are examples, not a recommendation to adopt one vendor.

S10. NKBA Planning Guidelines: Regional Boundary

https://nkba.org/planning-guidelines/

The public overview describes guidance intended for US/Canadian design practice and its code context. Only the overview was consulted, not the paid handbook. Do not label NKBA dimensions as German/EU requirements or cite handbook clauses not actually verified.

kitchen-planning-fr: Full Skill Definition

Source file: `.github/skills/kitchen-planning-fr/SKILL.md`

SHA-256: `c0b5ef0996f42b012de3aec18f350407b053b6c56890794de82d055810f9d5ac`

```
---
name: kitchen-planning-fr
description: "Plan or review a residential kitchen in France, from the measured brief to a proposal for professi
onal order review. Use for French kitchen layouts, cuisine amenagee or equipee, NF C 15-100 planning checks, VMC
 and hood coordination, copropriete renovation constraints, furnished-rental equipment, sink/preparation/hob pla
cement, dishwasher unloading, and cabinet or drawer allocation. France-first, metric guidance that distinguishes
 applicable requirements from ergonomic recommendations and household preferences. Not installation certificatio
n or a commercial-kitchen design standard."
---
```

Kitchen Planning for France

Purpose and scope

Produce a usable, measurable residential kitchen proposal for a project in France. Default to millimetres and EUR; confirm territory, budget basis, building type, tenure, and work scope before applying requirements. Respond in the user's language; keep French trade terms when they improve coordination.

This is a self-contained specialization of the workspace kitchen-planning approach. It does not require loading the Germany/EU skill. Use the planning reference for detailed checks, evidence qualifications, and worked examples.

The source catalog is optional. Load it only to verify a claim, provide citations, or update guidance. Do not load the bibliography merely to resolve source IDs: operational qualifications are already in this skill and the planning reference.

What is specifically French?

There is no single "normal European kitchen" or uniform French kitchen style. French-market layout guidance uses the same preparation, cleaning, cooking, and point-of-use storage principles as other European guidance (FR-S5).

Planning topic	France-specific adaptation
Electrical installation	Establish the applicable NF C 15-100 edition and work scope; distinguish ordinary kitchen sockets from specialized appliance circuits.
Ventilation	Identify the actual natural ventilation or VMC system, collective or individual, before choosing a hood or moving a cooking zone.
Renovation authority	Ask about copropriete, landlord consent, common services, structure, and exterior alterations; do not treat a kitchen seller's drawing as permission.
Rental brief	Distinguish owner occupation, location nue, furnished principal-residence lease, and tourist accommodation; their equipment and legal scopes are not interchangeable.
Accessibility	Determine applicable French housing provisions and user needs. Do not relabel German DIN dimensions as French compliance.
Procurement	Resolve cuisine amenagee/equipee labels into an itemized equipment and works schedule, a devis, and an installed TTC budget.
Lifestyle	Ask about actual cooking, meals, pantry and serving needs. Gas, wine storage, an island, rustic fronts, and elaborate cooking are not national requirements.

These topics are not all unique to France; their governing documents, terminology, and project decisions are localized here.

Authority and evidence

- Classify constraints as `regulatory`, `manufacturer`, `site`, `user_must`, `recommendation`, or `preference`.
- The first four are hard constraints when applicable. Recommendations and preferences are soft and may be traded off explicitly.
- A published standard is not automatically a universal retroactive legal obligation. Establish the applicable regulation, standard edition, contract, project dates, and scope with the responsible professional.
- A manufacturer summary is not the full standard. A product-specific installation requirement is different from that manufacturer's general planning advice.
- Never convert an unknown to a pass. If requirements conflict, stop the affected decision and obtain clarification; do not select the convenient rule.
- Photographs, scans, inferred dimensions, catalog nominal widths, and AI drawings do not replace a site survey or exact product documentation.
- The planner's geometry, catalog, prices, and accepted design remain authoritative. Propose reversible edits and obtain acceptance before changing the design.
- Do not claim electrical, gas, structural, ventilation, accessibility, rental, or installation certification. Do not authorize purchases or works.

Workflow

1. Establish the French project brief

Collect only the missing facts needed for the next decision; allow assumptions for concepts, clearly marked.

- Commune and territory: metropolitan France or the relevant overseas territory; do not silently apply metropolitan rules overseas.
- Apartment or house, approximate construction date, new build or existing dwelling, occupied or vacant during works.
- Owner or tenant; copropriete and syndic contact where relevant; private versus shared walls, ducts and service risers.
- Cosmetic replacement, partial technical renovation, complete electrical renovation, kitchen relocation, or structural alteration; project and permit/contract dates if relevant.
- Owner occupation, location nue, location meublee as a principal residence, bail mobilite, or tourist/secondary-residence use.
- Household size, simultaneous cooks, children, reach and mobility needs, wheelchair or seated work, and any applicable accessibility brief.
- Cooking frequency, appliance inventory, shopping/storage quantities, dining location, entertaining, and cleaning habits.
- Total installed budget in EUR TTC, retained items, time constraints, and responsibility for each trade.

Ask whether the household needs a table/coin repas, separate dining area, cellier, baking area, coffee station, large serving dishes, or bottle storage. Derive dimensions from their inventory and routines, not nationality.

2. Survey the room and services

Record wall lengths at relevant heights, diagonals, finished floor levels, ceiling height, wall build-ups, out-of-square corners, windows and opening arcs, doors, radiators, obstructions, and structural uncertainty.

Locate water, waste, isolation valves, electrical outlets and known circuit identities, gas if present, ventilation terminals, air inlets, ducts, and service access. Record delivery access, stairs, lift restrictions, and worktop handling constraints.

For each fact capture `value`, `unit`, `source`, `verification_state`, and measurement uncertainty. Use `unknown`, `inferred`, `user_verified`, or `professionally_verified`. Measure service positions from a declared datum and finished floor; do not infer concealed routes from visible endpoints.

3. Build the constraint register

Each record needs:

```
id
class
applicability
requirement
measurement_definition
source
status: pass | fail | unknown | not_applicable
affected_items
remedy_or_next_action
```

Run the French context checks in the reference before freezing a layout. Identify the person responsible for resolving each technical unknown.

- Electrical: confirm applicable edition and renovation scope with an electrician; reserve ordinary outlets separately from appliance connections. Socket count alone cannot establish compliance.
- Ventilation: a hood does not replace general dwelling ventilation (FR-S3). Preserve air inlets, transfer paths and extraction terminals. A visible communal shaft is not an approved powered-hood connection.
- Permissions: flag changes affecting shared services, structure, exterior outlets, or rented premises for the relevant professional, landlord, syndic, and authority checks before works.
- Rental: for a furnished principal-residence lease, check the scoped equipment list in the reference (FR-S4). Do not apply it to every kitchen or treat it as a complete rental-compliance assessment.
- Accessibility: establish applicable French requirements separately from ergonomic targets; unresolved legal applicability blocks a compliance claim, not an explicitly provisional concept.

4. Arrange five functional zones

Map food storage, preparation, cooking, cleaning, and clean storage/serving to real tasks. Prefer sink -> clear preparation surface -> hob, or its mirror, where the room allows it.

Provide a continuous useful preparation area, not a nominal length occupied by a drainer, appliance, corner, or landing zone. As an ergonomic starting point, aim for about 900 mm clear width; around 800 mm is a lower recommendation from manufacturer planning guidance, not a French statutory minimum (FR-S6). Check actual tasks and product cutout/edge requirements separately.

Do not invent a universal French 600 mm sink-to-hob prohibition. Assess splashing, heat, pan handles, worktop bridges, adjacent materials, and exact product requirements. A compact compliant layout can still have poor preparation space.

Keep grocery-to-fridge, washing-to-preparation, cooking-to-serving, and dirty-dish routes short and unobstructed. A triangle is an optional analytical aid, not a rule; a linear kitchen can work well. Separate through-traffic from hot work where feasible.

5. Design dishwasher unloading and storage together

Start with the dishwasher near the sink, subject to verified connection limits and usable access. Decide which side by actual unloading and loading tests, not handedness alone.

Test all of these together: dishwasher door open, racks extended, receiving drawer or cupboard open, and a person standing and turning. Do not make the user step over the door or balance dishes while opening storage.

Put everyday plates, bowls and dining cutlery close to a safe unloading position and the serving route. Put cooking utensils, pans and preparation tools near their use. Check that the shortest storage route remains accessible with doors open. A slightly longer unobstructed route can be better than blocked adjacent storage.

6. Allocate cabinets and drawers by contents

Create a schedule with cabinet ID, location, contents, quantities, largest item size, usable internal dimensions, load limit, opening mechanism, access frequency, and service conflicts.

- Use accessible drawers or pull-outs for frequently handled base storage where appropriate; check heavy dish stacks and hardware loads (FR-S7).
- Separate dining cutlery from cooking utensils. Give pans, lids, boards and trays usable storage near preparation/cooking.
- Size pantry storage to shopping habits; a cellier supplements but does not automatically replace daily point-of-use storage.
- Provide safe storage for knives and cleaning products, and a waste-sorting arrangement matching local collection and the household's carrying route.
- Keep shutoffs, traps, filters and appliance removal paths accessible. Under-sink space is not an empty rectangular cabinet volume.
- Use exact cabinet/appliance combinations: nominal 600 mm widths do not prove integration, ventilation, hinge, front, plinth or worktop compatibility.

7. Compare feasible alternatives

Normally compare two plausible layouts, such as a service-preserving option and a user-priority option. Include an island only if occupied circulation, services, budget and work surfaces support it.

Test grocery unloading, preparing a meal, draining a hot pan, dishwasher loading/unloading, serving, waste removal, and two people sharing the room. Include chairs pulled out, doors and drawers open, and a person at the oven. Test task lighting, glare, extraction noise and sightlines for an open kitchen.

Use 1000-1200 mm single-cook and 1200-1500 mm shared working aisles only as concept heuristics between the nearest closed projections, not French legal or wheelchair clearances. Then test actual open-state and occupied clearances; apply verified project requirements instead where stricter.

For repeated tasks, estimate:

```
daily_travel_m = sum(task_cycles_per_day *
                     sum(trip_count_per_leg * walkable_leg_length_m))
```

Count return trips explicitly. State assumptions; compare bending, awkward reaches, handling and collision risks separately. Do not present this proxy as a validated human-performance score.

8. Resolve the devis and installation boundaries

Translate cuisine amenagee/equipee into explicit inclusions, never assumed legal definitions. List meubles, facades, fileurs, joues, plinthes, plan de travail, credence, evier, robinetterie, electromenager, and fittings as applicable.

Separate supply, delivery, carrying, removal/disposal, preparation works, plumbing, electrical/gas work, ventilation, fitting, worktop templating/cutouts, finishing, testing and handover. Record exclusions, tax basis, quote validity, allowances and who owns final measurements. Do not promise a reduced VAT rate or a fixed total without a verified quote.

Ask the cuisiniste and trades to resolve interfaces before final ordering: sockets behind drawers, sink/trap versus bin, dishwasher front versus plinth, duct versus wall unit, and opening clearances versus fileurs are common coordination risks.

9. Deliver and gate readiness

Provide the brief and assumptions, dimensioned plan with opening envelopes, zone/task rationale, cabinet contents schedule, appliance/service schedule, constraint register, alternatives and trade-offs, installed-cost scope, and professional questions/next actions.

Use one readiness status:

- `concept`: assumptions or unresolved geometry/services remain; no order-ready implication.
- `dimensionally_checked`: stated geometry and operating envelopes checked against identified measurements and product data; list unresolved regulatory/technical matters separately.
- `ready_for_professional_order_review`: required survey, exact product compatibility, service feasibility, applicable permissions, and hard constraints have supporting evidence with no unresolved failures or unknowns. This still requires professional order review and is not installation approval or purchasing authorization.

Never call a proposal "French-compliant" solely because it follows this skill.

kitchen-planning-fr: Full Planning Reference

Source file: `.github/skills/kitchen-planning-fr/references/planning-reference.md`

SHA-256: `2d0713f9fb9cc12fdb24961131785341bc1388fbc100f8db20ab2bbc0ba52ac3`

France Kitchen Planning Reference

Operational reference for kitchen-planning-fr. Source IDs support traceability without requiring bibliography loading. Open the optional source catalog only for verification, citations or maintenance.

Evidence boundaries

Research reviewed on 2026-10-09. These are planning checks, not a complete French legal or installation specification. Refresh time-sensitive claims for the actual project. Full NF C 15-100 text, current project-specific accessibility provisions, and product-specific manuals were not inspected in this research.

French-market planning guidance supports familiar workflow principles, not a separate national geometry (FR-S5). Do not infer household habits, a required style, or a gas preference from the country. Imported appliances and retained furniture need exact compatibility checks, not assumptions based on EU marketing.

French technical and project gates

Electrical scope and layout reservation

Legrand's professional guide describes the NF C 15-100 series for new or entirely renovated low-voltage installations in France. It reports the 2024 revision, publication on 2024-08-23, and a 12-month overlap with the previous edition ending 2025-08-23; project reference dates include the contract, permit application or prior declaration (FR-S1). Schneider independently confirms a 2024 revision (FR-S2). These summaries do not determine a particular project's legal or contractual obligations.

Ask the electrician to identify the applicable edition, reference date, affected circuits, work scope, protection and testing requirements. Do not assert that replacing cabinets, selling a dwelling, or changing one appliance automatically requires complete rewiring. Conversely, do not use "existing installation" as an exemption from addressing identified hazards or new-work requirements.

Use these source-qualified reservations, not automatic compliance tests:

Situation	Reservation and qualification
Kitchen clearly greater than 4 m2	Legrand reports at least six ordinary sockets on a dedicated kitchen circuit, including four above the worktop. Identify appliance connections separately (FR-S1).
Kitchen clearly less than 4 m2	Legrand reports at least three ordinary sockets. Confirm applicable exception and placement with the electrician; reduced count is not proof of functional adequacy (FR-S1).
Exactly 4 m2, uncertain measurement, or open kitchen	The inspected web summary uses less-than/greater-than wording without resolving equality or area definition. Reserve six provisionally and keep applicability unknown until the electrician verifies the full rule. Do not encode a legal threshold from this ambiguity.
Hood outlet	Legrand describes an optional additional hood outlet at least 1800 mm above finished floor. This is an outlet position, NOT the hood-to-hob installation clearance (FR-S1). Confirm exact permitted position and product instructions.
Hob, oven, dishwasher and other substantial appliances	Record each model, rated load, connection method, required specialized circuit and isolation access. Have the electrician size and verify the installation. Ordinary socket count is not an appliance-circuit allowance.

Situation	Reservation and qualification
Appliance operation	Verify available supply and simultaneous loads, including retained appliances. Do not assume an existing socket or adapter makes an imported appliance suitable.

As a conservative planning rule, keep ordinary sockets and trailing leads away from sink and cooking surfaces and do not propose ordinary sockets directly above them. Have the electrician verify exact restrictions and hood exceptions against the applicable standard; this research did not independently verify the full placement clause. Do not import bathroom electrical volumes around a kitchen sink or assert an unsupported universal 600 mm sink/hob separation.

Coordinate outlet positions with credence joints, wall units, drawer backs, appliance depth and accessibility. Do not supply DIY conductor, breaker or wiring instructions as part of a furniture plan. Add task lighting without assuming that a ceiling point alone lights preparation adequately.

Ventilation, extraction and gas

A hood treats cooking emissions intermittently; dwelling ventilation renews indoor air more generally. Sauter's FAQ explains why a hood cannot replace VMC (FR-S3). This distinction does not prove that every existing French dwelling must receive the same mechanical system.

Survey natural ventilation, individual VMC, collective VMC, air inlets, transfer paths, kitchen extraction terminal, and any combustion appliance. Obtain the relevant design and maintenance information. Do not obstruct vents with cabinets or close a required air path during replacement works.

- Recirculation filters return air to the room; do not count them as outdoor exhaust or as a substitute for moisture removal by dwelling ventilation.
- Dedicated outdoor extraction requires a feasible, authorized duct route and coordinated air supply. Confirm flow, pressure loss, noise, discharge, fire/acoustic provisions and service access professionally.
- Do not connect a powered hood to an unidentified communal duct or VMC terminal. Treat compatibility and permission as unresolved until the responsible ventilation professional and building representative confirm the design. No universal connection prohibition is claimed from the inspected sources.
- Where gas or another combustion appliance is present, obtain professional review of air supply and depressurization risks before changing extraction. Do not infer safety from a closed cabinet drawing.
- Product manuals determine hood clearance, duct arrangement, filters, maintenance and compatibility; generic advice is not installation evidence.

Copropriete, rented premises and works

These are due-diligence gates, not a sourced statement of voting thresholds or permit law. Determine ownership and permissions before relocating services, cutting walls, changing collective ventilation, penetrating floors, or adding a facade/roof outlet.

Record the reglement de copropriete, syndic response, any required authorization, landlord agreement where relevant, structural assessment, and any mairie/planning-authority requirement. A verbal sales assurance does not resolve these dependencies. Identify who verifies the applicable rules and what evidence is still missing.

For an older building, ask the responsible professional whether surveys for hazardous materials or other pre-work investigations are required before invasive works. Do not diagnose materials from a photograph.

Rental equipment branch

Service Public's furnished-equipment page, verified 2026-03-06, covers a furnished dwelling used as the tenant's principal residence, including bail mobilite (FR-S4). Its kitchen/meal-related minimum equipment includes:

- Cooking plates.

- An oven OR a microwave, not necessarily both.
- A refrigerator and a freezer, OR a refrigerator with a freezing compartment reaching a maximum temperature of -6 C.
- Dishes sufficient for occupants' meals and cooking utensils.
- A table and seats, storage shelves, and lighting.

The full furnished-dwelling list also covers bedding, bedroom shading and appropriate cleaning equipment. An inventory and detailed furniture condition record accompany the lease. Kitchen planning alone cannot certify the whole dwelling.

Do not add a dishwasher to this minimum list. Do not require luxurious fitted cabinets or apply this list to an owner-occupied home, location nue, or tourist rental. For those branches, identify the applicable requirements separately. The decency overview reviewed during research distinguishes metropolitan and overseas contexts but its detailed equipment sections were not inspected; do not infer that unfurnished rental means "no kitchen obligations."

In a relevant furnished-rental project, separate `regulatory` minimum equipment from `user_must` additions and `preference` upgrades. Verify the current applicable legal basis before final review.

Accessibility

Confirm whether French housing accessibility provisions apply to the building category, new/existing context, project date, dwelling status and work scope. Ask the responsible professional to supply current requirements and their measurement definitions, including any accessible/evolutive provisions relevant to the project.

The official-partner accessibility portal distinguishes new housing, existing housing and building types, but the inspected page was last updated in 2021 and warns of CCH recodification (FR-S8). It is orientation only, not current numerical kitchen-compliance evidence.

Do not substitute German DIN, US guidance or a generic 1500 mm aisle for the applicable French rule. Separately test the actual user's turning, approach, knee/toe space, reach, handles, controls, sink protection, seated preparation and transfer needs. A user trial supplements, not replaces, applicable requirements.

Working dimensions and measurement definitions

Item	Concept starting point	Definition and authority
Main preparation width	About 900 mm; around 800 mm as a lower ergonomic recommendation	Clear usable worktop between interfering features; manufacturer planning guidance, not French law (FR-S6).
Single-cook working aisle	1000-1200 mm	Nearest closed projections, including handles; inherited planning heuristic, then test occupied/open states.
Shared working aisle	1200-1500 mm	Same measurement; concept heuristic, not an accessibility standard.
Worktop height	User/task-specific	Trial with actual users, sink depth and hob/pan height; do not impose a fixed "French height."
Appliance bay and ventilation	Exact selected system	Installation drawings, required air paths, fronts, fixings and service voids; never nominal cabinet width alone.
Sink/hob separation and worktop bridges	Product and task-specific	Distinguish clear preparation width from cutout-edge distances and structural bridges.
Accessibility dimensions	Project-specific, professionally verified	State current source, applicability and exact geometry before marking pass.

Constraint checklist

Create project records from these categories; do not automatically mark the checklist passed. Applicable regulatory, manufacturer, site and user-must requirements are hard constraints. Record unknowns and their owners.

ID	Check and evidence needed
FR-CTX-01	Territory, building type, tenure, lease branch and project dates established.
FR-CTX-02	Scope of works and applicable technical/regulatory documents identified by responsible professionals.
FR-PER-01	Shared property, structure, facade and communal-service alterations identified; permissions resolved where required.
FR-PER-02	Landlord and project-specific pre-work survey/authority dependencies resolved where applicable.
FR-GEO-01	Survey datums, finished surfaces, diagonals, tolerances and uncertainty recorded.
FR-GEO-02	Window/door arcs, radiators, ceiling and service obstructions represented.
FR-GEO-03	Delivery, lift/stair access, worktop transport and assembly path verified.
FR-FIT-01	Cabinet runs include fileurs, joues, fronts, handles, plinths and installation tolerances.
FR-FIT-02	Exact appliance niches, hinges, ventilation, front weights and door/plinth compatibility checked.
FR-FIT-03	Drawers, corner mechanisms, appliances and neighboring doors tested together.
FR-FIT-04	Walls/floors, cabinet fixings, suspended loads and worktop support verified.
FR-WAT-01	Water, drain route, levels and proposed relocation feasible; not inferred from endpoints.
FR-WAT-02	Valves, trap, bins, sink bowl and drawer envelopes compatible and accessible.
FR-WAT-03	Dishwasher hose routing, limits and connection access match selected product.
FR-ELE-01	Electrician confirms edition, reference date, work scope and applicable requirements.
FR-ELE-02	Ordinary socket count, area interpretation and usable placement verified separately from appliance circuits.
FR-ELE-03	Appliance loads, circuit provisions, supply, protection and isolation access professionally checked.
FR-ELE-04	Outlet/lead positions coordinated with wet/hot areas, furniture and credence; lighting supports tasks.
FR-VEN-01	Existing ventilation type, inlets, transfer paths and extraction terminals documented and preserved.
FR-VEN-02	Hood mode, duct feasibility, air supply, collective-system compatibility and permissions verified.
FR-VEN-03	Hood/product clearances, noise, filters and maintenance access checked.
FR-GAS-01	Gas/combustion installation and extraction interaction professionally assessed where applicable.
FR-TOP-01	Worktop cutouts, bridges, edge distances, joints and supports meet supplier requirements.
FR-TOP-02	Splash/heat exposure and adjacent surface/material compatibility resolved.
FR-USE-01	Continuous preparation surface remains usable with everyday appliances in place.
FR-USE-02	Dishwasher unloading tested with racks, receiving storage and user present.
FR-USE-03	Cabinet allocation reflects real contents, usable dimensions, loads and frequency.
FR-USE-04	Grocery, cooking, serving and waste routes tested without unsafe crossings.
FR-USE-05	Occupied chairs, multiple cooks and open oven/drawer states preserve required passage.
FR-ACC-01	Applicable French accessibility requirements and actual user needs both addressed.
FR-SAF-01	Hot-pan landing, child safety, knife/chemical storage and stable furniture addressed.
FR-SRV-01	Appliance removal, shutoffs, filters and repair access remain possible.
FR-REN-01	Correct lease/territory branch selected; relevant equipment inventory verified without overextending its scope.

ID	Check and evidence needed
FR-ORD-01	Itemized supply/work/exclusion schedule and installed EUR TTC budget agreed.
FR-ORD-02	Final measurements, exact products, technical interfaces, evidence and professional-review responsibilities resolved.

Worked checks

Open-state dishwasher conflict

A 1200 mm closed-front aisle looks generous. A dishwasher projects 600 mm into it and an opposing receiving drawer projects 500 mm. The residual gap is:

```
1200 - 600 - 500 = 100 mm
```

That is not usable standing/unloading space. It fails this simultaneous-use scenario even if neither closed cabinet footprint overlaps. Move the receiving storage, change opening direction/type or redesign the arrangement; then rerun the test with actual products and the user envelope.

Unloading travel

Assume four equal batches carried from dishwasher to storage, three return trips, and start at the dishwasher/end at storage. For illustrative measured walkable routes of 1.0 m and 2.5 m:

```
near_storage_m = (4 + 3) * 1.0 = 7.0
far_storage_m = (4 + 3) * 2.5 = 17.5
difference_per_cycle_m = 17.5 - 7.0 = 10.5
```

Do not use straight-line cabinet distances through open doors. The shorter route wins only if it remains safe and usable; bending and load handling need separate assessment.

French socket-count boundary

A kitchen is measured at 4.0 m2 with uncertainty, and the seller proposes three ordinary sockets. The inspected manufacturer's less-than/greater-than summary does not settle the boundary. Reserve six provisionally, ask the electrician for the applicable area/count rule, and keep that requirement `unknown`. Even six visible sockets cannot prove circuit or placement compliance.

Furnished-rental distinction

A principal-residence furnished letting has a hob, microwave and suitable refrigerator/freezing compartment but no conventional oven or dishwasher. The oven-or-microwave requirement is not failed merely because there is no conventional oven; the dishwasher is not on the inspected minimum list. Verify the rest of the equipment, lease scope and dwelling requirements before any compliance conclusion (FR-S4).

Shared-duct constraint beats preferred island

A proposed island hood depends on an unidentified communal shaft. Do not draw a connection and mark ventilation passed. Keep feasibility/permission `unknown`; compare a service-preserving arrangement and a professionally assessed recirculation option that retains general ventilation. Neither becomes order-review-ready until all applicable hard constraints are resolved.

Final review questions

Can a user prepare, cook, unload and serve with real occupied/open states? Are French project requirements evidenced rather than presumed? Can installers resolve every service/furniture interface from the documents? Are remaining assumptions visible and incompatible with any claimed readiness? If not, return the proposal to the appropriate earlier stage.

kitchen-planning-fr: Full Source Catalog

Source file: `.github/skills/kitchen-planning-fr/references/source-catalog.md`

SHA-256: `b13943330653ae519601a8c5a9f438b44861d569a026c2bf99da42c515c73d8d`

France Kitchen Planning Source Catalog

Optional research/provenance file for the skill and its operational reference. Do not load during routine planning unless a claim needs verification, the user requests citations, or guidance is being updated.

Reviewed on 2026-10-09. Source claims are paraphrased. A retrieval date is not a guarantee of current legal applicability. Commercial guidance is identified as such; general manufacturer advice is not a selected product's installation manual.

FR-S1: Legrand, NF C 15-100 professional guide

- URL: https://www.legrand.fr/pro/normes-et-reglementations/norme-nf-c-15-100/norme-nf-c-15-100-suivez-le-guide
- Publisher/type: Legrand, manufacturer technical summary; page headed "Norme NF C 15-100 en 2026 : on vous dit tout."
- Inspected: introduction, structure of the standard, circuits prises, and related scope text.
- Supports: 2024 revision and reported transition dates; new/entirely renovated installation context; ordinary kitchen socket reservations; separate specialized appliance connections; four worktop sockets within the six-socket case; optional additional hood outlet at 1800 mm minimum above finished floor.
- Limits: not the complete standard or an independent legal interpretation. The page uses `< 4 m2` and `> 4 m2` without explicitly resolving equality. Do not convert that gap into a legal rule. Its general appliance circuit examples are not a complete design for any particular supply or appliance inventory. Full placement clauses were not independently inspected.
- Use in skill: French electrical intake, reservations, electrician handoff, and boundary-case test. No DIY wiring instructions or automatic retrofit obligation.

FR-S2: Schneider Electric, NF C 15-100 overview

- URL: https://www.se.com/fr/fr/work/support/local/reglementation/norme-nfc15-100/
- Publisher/type: Schneider Electric France, manufacturer standards overview.
- Supports: identification of the 2024 revision and availability of updated guidance; AFNOR standard versus explanatory guide distinction.
- Limits: detailed expandable answers and the downloadable guide were not inspected. Does not independently verify the numeric kitchen reservations used from FR-S1.
- Use in skill: cross-check of the revision context, not project compliance evidence.

FR-S3: Sauter, hood versus VMC

- URL: https://www.confort-sauter.com/aide/vmc/installation/ma-hotte-de-cuisine-peut-elle-remplacer-une-vmc/
- Publisher/type: Sauter, manufacturer ventilation FAQ.
- Supports: continuous air renewal by VMC versus intermittent removal of cooking emissions by a hood; their complementary functions.
- Limits: blanket marketing statements about mandatory VMC are not used as universal retrofit law. This FAQ did not establish legal permission or a universal prohibition for powered-hood connection to a communal system.

- Use in skill: retain general ventilation, investigate the actual system, and require professional compatibility/permission evidence.

FR-S4: Service Public, furnished-dwelling equipment

- URL: https://www.service-public.gouv.fr/particuliers/vosdroits/F34769
- Title: "Bail d'habitation : qu'est-ce qu'un logement meuble ?"
- Publisher/type: French public-service administrative guidance; page verification date 2026-03-06.
- Supports: minimum equipment for a furnished principal-residence lease, including bail mobilite; cooking plates, oven OR microwave, refrigerator/freezer alternative, dishes, utensils, table/seats, storage and lighting; inventory documentation.
- Limits: not a universal specification for owner occupation, location nue, holiday rentals or every overseas jurisdiction. The whole dwelling has additional requirements beyond the kitchen. Verify the current applicable legal basis and territorial scope for the project.
- Use in skill: scoped rental branch and equipment distinction, not a complete decency certificate.

FR-S5: IKEA France, practical kitchen layout

- URL: https://www.ikea.com/fr/fr/rooms/kitchen/how-to/idees-dagencement-pratique-pour-la-cuisine-pub6d21e890/
- Title: "Idees d'agencement pratique pour la cuisine."
- Publisher/type: French-market retailer planning advice.
- Supports: preparation between sink and cooking, nearby dishwasher/waste, storage at point of use, and several layout forms.
- Limits: activity-triangle and equal-distance language is advisory, not mandatory. One retailer cannot establish the prevalence of national household habits.
- Use in skill: evidence that familiar ergonomic principles also apply in France; no invented French triangle or style requirement.

FR-S6: Blum, preparation storage and worktop

- URL: https://www.blum-inspirations.com/de-eu/wissen/speisenvorbereitung-leicht-gemacht-der-passende-schrank-unterstuetzt-sie-dabei
- Publisher/type: Blum, German-language European manufacturer ergonomics/planning advice, reused from the general kitchen research.
- Supports: preparation worktop and related storage; approximately 900 mm recommended working width and around 800 mm lower ergonomic recommendation.
- Limits: not French regulation and not a universal minimum. Usable clear surface, tasks, products and users matter more than a nominal run length.
- Use in skill: explicitly labeled transferable ergonomic recommendation, not country-specific compliance.

FR-S7: Blum, accessible dish storage

- URL: https://www.blum-inspirations.com/de-eu/wissen/geschirr-perfekt-organisiert-im-unterschrank-mit-auszuegen
- Publisher/type: Blum, German-language European manufacturer storage guidance, reused from the general kitchen research.
- Supports: accessible base-unit pull-outs for dishes and handling-oriented storage allocation.
- Limits: commercial guidance; verify actual usable dimensions, loads and user reach. Does not prove that every drawer is suitable for every dish stack.
- Use in skill: contents-driven drawers and dishwasher-unloading evaluation.

FR-S8: Accessibility portal, housing applicability orientation

- URL: https://www.accessibilite-batiment.fr/logements-neufs/generalites-et-definitions/code-de-la-construction-et-de-lhabitation.html
- Publisher/type: DHUP-CSTB-Cerema accessibility portal, regulatory orientation.
- Inspected: general distinctions for collective housing and individual houses; page update 2021-09-10 and warning concerning CCH recodification from 2021-07-01.
- Supports: the need to distinguish building categories, project context and applicable documents rather than treating every kitchen identically.
- Limits: old numbering and dated material; not relied on for a current legal determination or any French kitchen dimension. Current full applicable provisions must be obtained for the project.
- Use in skill: accessibility applicability gate; no imported DIN or unverified French numeric requirement.

Research exclusions and maintenance

- Service Public's rental-decency overview at https://www.service-public.gouv.fr/particuliers/vosdroits/F2042 was only partially accessible. Its metropolitan/overseas distinction informed the intake; detailed kitchen-equipment provisions were not verified.
- IZI by EDF's kitchen-electrical article was inspected but not adopted as authority because its broad claims and comments contained ambiguities or conflicts. In particular, no universal 600 mm sink/hob rule, kitchen bathroom-like volumes, mandatory kitchen RJ45, or automatic whole-dwelling upgrade on sale is adopted.
- IKEA France's hood article was inspected but is not used for generic hood heights, filter lifetimes or claims that recirculation removes humidity outdoors.
- Primary ventilation legislation could not be retrieved from Legifrance during research. No numeric statutory ventilation rate or categorical communal-duct legal rule is asserted.
- Copropriete permissions, landlord agreement, quotations and trade coordination are practical due-diligence checks. Voting thresholds, permit procedures, statutory quote requirements and VAT rates were not established and are not prescribed.
- No primary text or current project-specific French accessibility clause was verified. Do not turn portal orientation into an accessibility certificate.
- Before updating a numeric requirement, check the full current source, territorial/project applicability, exceptions and measurement definition; update the operational guidance and source qualification together.

living-room-planning: Full Skill Definition

Source file: `.github/skills/living-room-planning/SKILL.md`

SHA-256: `d19ab7eb87f0226527292d255f977180e04be16a230ccb24c995d89f64627a66`

```
---
name: living-room-planning
description: "Plan or review a living room, lounge, sitting room, or Wohnzimmer using household activities, veri
fied geometry, and practical furniture constraints. Use when arranging sofas, armchairs, recliners, sofa beds, c
offee tables, TV/media units, bookshelves, or storage; comparing conversation and TV layouts; checking circulati
on, accessibility, lighting, glare, child safety, or open-plan living zones. Defaults to Germany/EU context and
metric units; separates requirements from ergonomic recommendations and preferences. Includes operational checks
 and an optional research source catalog."
---
```

Living Room Planning

Purpose and loading policy

Create practical, explainable living-room proposals that work for the household's real activities. Optimize comfort, access, storage, and ease of changing activities before decorative composition or maximum furniture count.

Default to Germany/EU context, dimensions in millimetres, and costs in EUR unless the user specifies otherwise. Confirm the actual country and, where relevant, German federal state. There is no single set of EU living-room layout rules.

- Read the planning reference before applying numerical guidance, assessing accessibility or installation risks, or declaring a layout dimensionally checked.
- Read the optional source catalog only to verify a source, answer provenance/citation questions, resolve disputed claims, or update guidance.
- Do not load the source catalog merely to resolve source IDs during routine planning. The operational reference contains the qualifications needed for normal use.
- This skill supports planning, not structural design, electrical installation, fire-safety approval, clinical assessment, or permission to purchase.

Non-negotiable discipline

1. Distinguish measured facts, assumptions, user preferences, recommendations, and mandatory requirements.
2. Never infer exact room dimensions, wall construction, load capacity, or safe clearances from a photograph alone.
3. Preserve the existing planner's authoritative geometry, product catalog, and accepted design. Proposals are reversible drafts until explicitly accepted.
4. Check occupied, moving, and fully deployed furniture, not just closed footprints. A sofa bed's sleeping surface is not its overall deployed size.
5. Do not treat a reachable-looking gap, generic furniture dimension, or aesthetic rule as proof of accessibility or safety.
6. Keep required exits and escape openings usable in every relevant mode. A blocked exit is not acceptable because guest use is infrequent.
7. Unknown is not pass. Missing critical dimensions or installation evidence must remain visible.
8. Do not prescribe a TV, fireplace, large sectional sofa, rug, or symmetry unless it serves the brief.

1. Establish the brief and survey

Ask only questions that can change the next decision; group missing information instead of sending a long questionnaire. Start with daily occupants, activities, existing furniture to retain, room geometry, and budget. If measurements are missing, offer a labelled concept and a concise measurement list.

Capture:

Subject	Required information
Household	Daily and visiting occupants; children; mobility aids; reach, vision, hearing, balance, transfer, and sensory needs; foreseeable changes
Activities	Conversation, TV/gaming, reading, music, play, exercise, hobbies, occasional work/dining/sleeping; priorities, frequency, duration, and simultaneous use
Geometry	Room outline, ceiling/slopes, columns, niches, skirtings, changes in level, actual door and window positions, sill heights, handles, opening arcs, thresholds
Connections	Entrance, adjacent rooms, kitchen/dining area, balcony/terrace, required exits and escape openings; routes used while carrying items
Services	Radiators, thermostats, ventilation openings, fireplace/stove and hearth, sockets, switches, data/TV points, ceiling outlets, fixed lighting
Daylight and sound	Window orientation, glare at typical viewing times, shading, street/neighbour noise, adjacent bedrooms, reflective surfaces
Products	Retained pieces and exact candidate models; external and usable internal dimensions, weight/load limits, open/deployed sizes, anchoring and ventilation instructions
Inventory	Books, games, toys, blankets, media devices, controllers, chargers, hobby/work items, guest bedding; quantities, sizes, weights, access frequency
Delivery and work	Building entrance, stairs, lift, door turns, package dimensions, assembly space, wall permissions, access for installers and later servicing
Preferences and cost	Must-keep items, comfort trials, style/materials, cleaning needs, budget including delivery/assembly, wiring, mounting, shading, lighting, and disposal

For each critical fact record `value`, `unit`, `source`, `verification_state`, and uncertainty/tolerance. Use `unknown`, `inferred`, `user_verified`, or `professionally_verified`. A product's nominal width does not prove the available room width.

Create an activity table with `mode`, `users`, `frequency`, `simultaneous_activities`, `required_items`, `deployed_state`, and `reset_steps`. Distinguish daily seats from occasional guest seats. Do not count the same floor area twice for simultaneous activities.

2. Build a constraint register

Use these classes:

- `regulatory`: verified, applicable local requirements; record jurisdiction and edition.
- `manufacturer`: exact product instructions, clearances, loads, mounting, and intended use.
- `site`: verified fixed geometry, services, structural limits, and permissions.
- `user_must`: explicit non-negotiable household requirements.
- `recommendation`: ergonomic or functional advice, including labelled planning heuristics.
- `preference`: negotiable visual or lifestyle choices.

The first four classes are hard constraints. The latter two are soft unless explicitly adopted as a project requirement. Identify safety hazards even when their legal classification is unverified; do not bury them in an aesthetic score. A user preference cannot waive a mandatory requirement.

Record `id`, `class`, `applicability`, `requirement`, `measurement_definition`, `source`, `status`, `affected_items`, and `remedy_or_next_action`. Status is `pass`, `fail`, `unknown`, or `not_applicable` with a reason. Include the room mode in each state-dependent check.

Resolve conflicting requirements explicitly. Reject hard failures from viable options; separate options needing missing evidence from checked options. Escalate wall loading, fixed electrics, combustion/hearth conditions, or accessibility compliance to the appropriate professional.

3. Reserve routes and activity zones

1. Mark fixed features, required access, opening arcs, and no-go areas before placing furniture.
2. Connect the entrance to used seats, storage, adjacent rooms, windows, and balcony along genuinely walkable paths.
3. Prefer through-routes outside the conversation group and TV sightline where geometry permits. Show the compromise when a through-room makes this impossible.
4. Size the main zone for daily activities; reserve secondary zones only where the brief requires them.
5. Test daylight and noisy/quiet activities together. A shelf or curtain can separate views without providing acoustic privacy.
6. Locate movable furniture's parking positions. A coffee table moved for a sofa bed must not be parked in a doorway or another required activity zone.

Possible strategies include facing/angled seating for conversation, sofa plus movable chairs for mixed use, a media-oriented group, or distinct quiet/active zones. These are alternatives, not mandatory room types.

4. Fit the seating to the users

- Select usable seat width, seat depth, compressed seat height, support, armrests, and ease of standing through user trials, not a universal ideal sofa size.
- Count seats that people can actually use together. A chaise, corner junction, ottoman, or armless end is not automatically an equivalent full-time seat.
- For conversation, arrange faces so users can hear and see each other without sustained twisting. Try angled chairs or more than one small group before enlarging a distant perimeter arrangement.
- Check entry to each seat, knees/feet, standing, transfers, walking-aid parking, swivel arcs, recliner mechanisms, and access to controls.
- Position drinks, a book, glasses, and remotes within comfortable reach. A side table can serve a seat better than a large central table.
- Treat coffee-table reach space and through-circulation as different things. An easy-to-reach table gap is not a corridor.
- Verify sofa-bed deployment, bed-making, access to bedding, night-time routes, and the effort/strength needed to operate it. Do not assume an occasional sofa bed is suitable for regular sleep.

5. Locate TV, media, and listening equipment

Only apply this step when relevant to the brief.

1. Identify primary and secondary viewing seats and measure actual seated eye positions.
2. Start with the screen centre near seated eye level, then trial the real posture, screen size, subtitles, and viewing comfort. Do not choose a fixed mounting height from the floor alone (S3, S4).
3. Choose viewing distance with screen size, resolution, content, eyesight, and preference. There is no universal distance multiplier suitable for every user (S5).
4. Check horizontal viewing positions, head turning, and sightline obstruction from every claimed TV seat. The panel's specified viewing angle is not a comfort guarantee.

5. Check reflections from windows and luminaires at typical times. Side-on placement to windows is a candidate, not an automatic glare solution; verify with the actual room and shading.
6. Check exact TV foot spacing/depth, furniture load capacity, anti-tip measures, bracket compatibility, wall substrate, and cable strain through mount movement. A wider cabinet alone does not prevent tipping (S3, S8).
7. Allow actual device ventilation, connector depth, plug removal, service access, and remote/signal operation behind doors. An opening for cables does not establish adequate cooling.
8. Position speakers using product guidance, listening trials, and neighbour considerations. Do not promise acoustic isolation from rugs, curtains, or decorative panels.

Do not default to a TV above a fireplace: both viewing posture and verified heat/installation limits must work. An unknown heat condition blocks mounting approval.

6. Assign contents before choosing storage

Contents	Preferred relationship	Operational check
Remotes, glasses, current book	At the relevant seat	Reach while seated; no risky leaning or trailing charger
Games and controllers	Near play/media area	Access and return without disconnecting equipment or clearing occupied seats
Toys	Near the agreed play area	Child access where appropriate; anchoring, lid/pinch risks, tidy-up route
Books and records	Near reading/listening area	Real shelf depth, load limits, adjustable spacing, safe retrieval
Throws and cushions	Near relaxation seats	Storage opens while people are seated; no need to empty the tabletop
Guest bedding	Near sofa bed	Retrieve before/during deployment; dry storage and manageable lifting
Work or hobby equipment	Near its use zone	Fast setup/reset, suitable work surface, secure storage where needed
Hazardous or fragile items	Controlled location	Access restrictions appropriate to household; height alone is not a lock

Use internal clear dimensions, hinges, drawer travel, shelf loads, and actual object sizes. Locate frequent items for easy retrieval; avoid heavy overhead storage. Compare closed storage for clutter control with open storage for access/display, without sacrificing equipment cooling.

Storage should reduce repeated interruptions: getting up, asking someone to move, reaching across a screen, unloading a tabletop to lift a lid, or shifting furniture. Shorter walking distance is useful only when access remains safe and convenient.

7. Integrate light, environment, and safety

- Combine general, task, and ambient/accent lighting with independently usable controls (S2). Reading needs suitable light on the page, not just a bright room.
- Check source glare, reflected screen glare, contrast, shadows, and seated/standing sightlines. Place switches where users can reach them on entry and at key seats; retain usable non-app control.
- Plan cables and outlets before accepting floating furniture. Do not hide extension leads under rugs or use cable routing that creates trip, pinch, or heat hazards. Fixed electrical alterations require qualified work.
- Keep radiator heat flow, thermostat sensing, and ventilation unobstructed (S7). Check exact installation instructions and any cold/damp wall concern; do not invent a universal furniture-to-wall gap.

- Treat required furniture/TV anchoring as part of the layout. Verify suitable wall/floor construction and fixings; an unknown wall is not a passed mounting check (S8).
- Address climbing access to windows, accessible blind cords, unstable tall furniture, loose rugs, sharp/glass edges, hot surfaces, and moving mechanisms. Prefer cordless window coverings around young children; moving furniture alone does not eliminate cord hazards (S9).
- For echo, consider appropriate absorptive furnishings subject to cleaning, trip, and fire constraints. Speech privacy, bass transmission, and sound isolation require different assessment; do not invent decibel improvements.
- If sustained work or dining is required, check a suitable table/chair/task-light setup. A sofa with a laptop tray is not evidence of an ergonomic full-time workstation.

8. Generate and test alternatives

Offer two or three meaningfully different layouts when useful, not multiple cosmetic versions. Preserve existing furniture when it meets the brief. Explain when only one credible option fits.

For each option test the required states:

- Normal day with the usual seats occupied.
- Conversation/guests with occasional seats placed and used.
- TV/gaming with sightlines, glare, and controller/device access.
- Reading/work/play occurring simultaneously where requested.
- Recliners, drawers, doors, swivel chairs, and sofa bed through their full motion and deployed states.
- Night-time access, window operation, cleaning, servicing, and eventual furniture removal.

Measure the narrowest clear route between actual obstacles, including occupied feet/knees and deployed parts. Check turning and approach geometry as well as width. Use tolerance-aware dimensions; marginal clearance with unverified geometry remains unknown.

Use scaled geometry checks where available and full-size tape/cardboard mock-ups with user trials where practical. Describe exactly what was checked; a language-model review alone is not deterministic collision validation.

Compare options by hard failures/unknowns first, then daily comfort, circulation, conversation/media suitability, storage access, reset effort, light/glare, maintenance, cost, and preferences. Let the user set priorities; do not invent authoritative scoring weights.

Count furniture moves, lifting, setup time, interruptions, and actual walkable trips for repeated routines. State assumptions and frequency. Do not optimize walking distance at the expense of comfort, inclusion, or required access.

9. Deliver the proposal

Provide:

1. Brief, priority activities, assumptions, and missing evidence.
2. Recommended layout and alternatives, using labelled walls or a defined coordinate origin; include external furniture dimensions and critical clearances.
3. Seating and storage assignments, TV/light relationships, services, and required anchors.
4. Mode-by-mode checks, worst pinch points, and the destination of every displaced item.
5. Constraint register with failures, unknowns, source type, and next action/owner.
6. Budget range, retained/new items, installation/delivery dependencies, and next measurements or trials.
7. An explicit readiness level.

Use `concept` when dimensions or products are provisional. Use `dimensionally_checked` only for the listed modes and checks supported by verified geometry/product envelopes. Use `ready_for_professional_order_review` only when critical unknowns and hard failures are resolved and delivery/installation evidence is assembled. None of these labels authorizes purchase or certifies regulatory compliance; professional review and user acceptance remain separate.

Counterexample quality checks

Before concluding, reject these false passes:

- The closed recliner leaves a wide route, but its footrest blocks the balcony when used.
- Six nominal sofa places exist, but two require twisting and one blocks access to another.
- A nearby storage ottoman is convenient only when nobody uses it and everything is removed from its lid.
- A sofa bed opens only after the coffee table is moved, with no safe parking place for that table.
- A low TV cabinet fits, but the television feet overhang or its anti-tip provision is unresolved.
- A 1500 mm square is shown somewhere in the room, but the wheelchair user cannot reach a seat, storage, or exit.
- A room divider hides a desk but does not make concurrent calls and television acoustically private.
- A layout photographs well but covers a radiator, blocks a window handle, or requires a cable across a walking route.

living-room-planning: Full Planning Reference

Source file: `.github/skills/living-room-planning/references/planning-reference.md`

SHA-256: `7b6ca91f49fb6827b861b147093b53172fcdf0b82aa4554fb9c711eefb0caa3e`

Living Room Planning Reference

Operational guidance for the living-room-planning skill. Defaults: Germany/EU context, millimetres, EUR. Source IDs identify research without requiring bibliography loading.

Load the optional source catalog only for source verification, citations/provenance, disputed claims, or maintenance. Do not load it just to resolve IDs in routine planning.

Evidence boundaries

- S1-S3 are IKEA consumer guidance on activity zones, lighting, and TV furniture. They support design reasoning, not building-code compliance or endorsement of particular products.
- S4-S5 are Sony general support guidance on TV height and distance. They are not exact product installation manuals or universal health limits.
- S6 is a secondary summary of DIN 18040-2 living/sleeping-room accessibility. Its index identifies the 2011-09 edition. Verify the current edition, project applicability, state adoption, and any contractual requirements before treating a dimension as mandatory. The full standard was not reviewed.
- S7 is German consumer advice supporting unobstructed radiator heat flow and thermostat sensing, not a numerical clearance standard.
- S8-S9 are US CPSC safety guidance on tip-over and window-covering cords. The hazard principles are relevant; US legal provisions are not German/EU requirements.
- The workflow, check catalog, and examples below are planning synthesis. A source supports only the stated claim, not every adjacent check. Project-specific manuals, structural conditions, and local requirements still need verification.

Dimensions and measurement definitions

The heuristic rows below are deliberately provisional starting points for mock-ups. They are not sourced standards, minimum legal dimensions, accessibility specifications, or guarantees of comfort. Override them for actual users, geometry, products, and applicable requirements.

Topic	Starting guidance	Measurement and qualification
Main walking route	Explore 900-1200 mm clear width	Skill heuristic for initial ordinary residential layouts; measure narrowest obstacle-to-obstacle gap in the relevant occupied mode. Passing, carrying, mobility aids, turns, or applicable rules may need more. This does not certify an escape route.
Seat to coffee table	Try 400-500 mm	Skill heuristic between seat front and table edge for reach and leg space; trial actual posture. Not a through-route or wheelchair-transfer clearance. Side tables or no central table may be better.
Conversation spacing	Trial roughly 1500-2500 mm between seated head positions	Skill heuristic for a small conversational group, not between sofa backs. Hearing, room noise, furniture angles, and personal space govern the result.

Topic	Starting guidance	Measurement and qualification
Turning/movement area	S6 reports 1200 x 1200 mm; 1500 x 1500 mm for unrestricted wheelchair use	Secondary DIN summary for relevant living/sleeping rooms. Distinguish general barrier-free from the R category; verify applicability and full provisions. A free square alone does not prove accessible use.
Space along/in front of furniture	S6 reports 900 mm depth; 1500 mm for unrestricted wheelchair use	Secondary summary; not a universal route-width substitute. Verify the actual approach, transfer, furniture function, and applicable category.
Seat dimensions	User- and product-specific	Measure usable width/depth and compressed seat height; trial support, feet placement, standing, and transfers. No fixed ideal dimensions specified here.
Screen centre height	Start near seated eye level (S3, S4)	Finished floor to centre of visible screen; use actual primary viewers and posture. Reclined viewing may require a different tested arrangement.
Viewing distance	No mandatory multiplier (S5)	Eye position to screen; note size, resolution, content, subtitles, and preference. Distinguish this from wall-to-sofa distance.
TV support/mount	Exact product dimensions and limits	Check TV feet, support depth, load rating, anchors, wall construction, bracket specification, and full movement envelope. Screen diagonal alone is insufficient.
Sofa bed/recliner	Exact fully deployed dimensions and motion	Include mechanisms, rear travel, user operation/standing, feet, bedding access, and displaced furniture. No generic clearance is approved.
Cabinet/device clearances	Exact manuals and usable internal space	Include plugs, cable bend space, drawer/door travel, ventilation, load capacity, and service removal.
Radiators, stove/hearth, cold walls	Site and product assessment	No universal gap. Keep heat flow and controls usable (S7); obtain verified combustion clearances and investigate moisture concerns.
Lighting	Activity-specific placement and controls (S2)	No mandatory domestic lux or colour-temperature target is established by this research. Agree needs; use photometric data/measurement for quantified claims.

Overlap of movement areas may be permissible in an applicable accessibility design, but do not overlap fixed obstacles or simultaneous occupied/deployed functions. Record exactly which states can share space and validate the applicable provisions.

Practical check catalog

These are check categories, not a claim that all checks apply or are legally required. Instantiate the relevant checks with the constraint class, measurement, mode, evidence, status, and next action. Preserve `unknown` where evidence is missing.

ID	Check	Failure or unresolved condition to expose
GEO-01	Survey and tolerances	Furniture accepted against estimated wall dimensions, unmeasured skirtings, slopes, or out-of-square corners
GEO-02	Openings	Door/window arcs, handles, curtains, thresholds, or escape openings obstructed
GEO-03	Route continuity	A nominal clear width exists locally but the full entrance-seat-storage-exit route fails
GEO-04	Required access	Balcony, services, ventilation, cleaning, or emergency access sacrificed to furniture
USE-01	Activity brief	Maximum seating or TV dominance chosen without household priorities
USE-02	Simultaneous use	Work, play, conversation, or dining claim the same space at the same time

ID	Check	Failure or unresolved condition to expose
USE-03	Real seating capacity	Corner/chaise/occasional places counted as equivalent comfortable daily seats
USE-04	Seated comfort	Unsupported feet, excessive seat depth, difficult standing, or inaccessible controls
USE-05	Conversation	Sustained twisting, blocked faces, distance/noise incompatible with hearing needs
USE-06	Reach and standing	Table placement forces leaning or removes space for feet, standing, or transfers
FIT-01	Actual external envelope	Product footprint excludes arms, legs, handles, overhang, or cables
FIT-02	Motion and deployment	Recliner, swivel chair, drawer, or sofa-bed travel collides with people or objects
FIT-03	Reset and parking	Displaced tables, cushions, bedding, or guest seats have no safe destination
FIT-04	Delivery and assembly	Package or assembled piece cannot negotiate entry, stairs, lift, or room turns
STO-01	Contents and loads	Object sizes/weights exceed usable shelf/drawer dimensions or ratings
STO-02	Retrieval	Frequent items require moving occupants, unloading a lid, or crossing active viewing
STO-03	Safe reach	Heavy overhead items, climbing, insecure hazardous contents, or inaccessible essentials
MED-01	TV eye geometry	Screen height/distance/angle chosen without the primary viewers and actual screen
MED-02	All claimed viewing seats	Obstructed picture, neck turning, subtitle difficulty, or unsuitable off-axis image
MED-03	Glare	Window or luminaire reflections untested at relevant times
MED-04	TV support and mounting	Feet overhang, inadequate loads, incompatible bracket, unknown substrate or fixing
MED-05	Device operation	Cooling, cable/plug space, service removal, or remote/signal access missing
LGT-01	General/task/ambient light	One source cannot serve agreed activities or leaves routes inadequately visible
LGT-02	Control and glare	Unreachable controls, app-only dependency, glare or harsh contrast at a used seat
ENV-01	Heating and ventilation	Sofa, cabinetry, or curtains obstruct radiator flow, thermostat sensing, or vents
ENV-02	Moisture and finishes	Cold/damp wall concern ignored; finishes/rugs unsuitable for actual conditions
ENV-03	Acoustic compatibility	Visual screen treated as sound isolation; no plan for concurrent noisy/quiet use
ACC-01	Accessible movement	Applicable clear areas, approaches, turning, and route continuity not verified
ACC-02	Inclusive use	Seat transfer, aid parking, reach, contrast, hearing, or control needs not met
SAFE-01	Stability and anchoring	Furniture, TV, or room divider lacks required stability/anchors and verified substrate
SAFE-02	Child/window safety	Climbing route, accessible blind cord, pinch/lid risk, or hazardous contents unresolved
SAFE-03	Trips and impact	Loose rug edges, trailing leads, low unseen obstacles, sharp/glass edges on routes
SAFE-04	Heat and combustion	Unverified stove/hearth/TV relationship, combustible furnishings, or obstructed provisions
SRV-01	Power and data	Unsafe extension routing, pinched leads, insufficient outlets, or unapproved fixed work
ORD-01	Ordering evidence	Product variants, prices, measurements, permissions, or installation checks unresolved
ORD-02	Readiness claim	Concept labelled installation-ready, unverified checks passed, or approval implied

Worked examples

All dimensions, timings, and counts in these examples are invented teaching inputs, not recommended product clearances or measured performance.

1. Closed-state route that fails when used

A straight passage between a closed recliner front and a cabinet front is 1250 mm. In the same cross-section, the footrest extends 550 mm into the passage and an opposing drawer extends 400 mm.

```
remaining_gap_mm = 1250 - 550 - 400 = 300
```

The positive gap does not imply a usable route. Feet, the person retrieving contents, and approach/turning space have not even been included. It fails an agreed 900 mm project route target. If this is the only access to an exit, rejecting the simultaneous state alone is not a sufficient remedy; relocate or resize furniture so required access remains usable in intended operation.

This subtraction is valid only for aligned projections at that cross-section. Offset furniture, curves, swing arcs, and room turns require actual geometry checks.

2. Sofa-bed mode includes its displaced furniture

A room has 3500 mm clear depth on the deployment axis. The bed's verified overall deployed depth is 2200 mm and the opposite cabinet is 400 mm deep.

```
nominal_foot_gap_mm = 3500 - 2200 - 400 = 900
```

This is not yet a pass: check doors, drawer access, operator standing, side approach, night-time route, and tolerances. A 600 mm deep coffee table parked in that gap leaves only 300 mm on the assumed aligned section. Find a separate safe parking position or change the furniture plan. Do not claim the mattress length is the deployed furniture depth.

3. Viewing geometry, not a prescribed distance

For a centred, perpendicular viewer and a 16:9 screen:

```
diagonal_mm = diagonal_inches * 25.4
screen_width_mm = diagonal_mm * 16 / sqrt(16^2 + 9^2)
horizontal_angle_degrees = 2 * atan(screen_width_mm / (2 * eye_distance_mm)) * 180 / pi
```

A 65-inch screen is about 1439 mm wide. At a 2500 mm perpendicular eye-to-screen distance it spans about 32.1 degrees horizontally. This describes the geometry; it is not a recommended angle, a safety threshold, a panel off-axis specification, or a comfort guarantee. Trial content, subtitle readability, eyesight, and actual seated posture (S4, S5).

4. Storage reduces reset effort, not just distance

Suppose daily hobby setup/reset requires moving a table twice and clearing an ottoman lid twice, each assumed to take 20 seconds. A nearby accessible drawer avoids those four actions without creating another obstruction.

```
avoided_actions_per_day = 2 + 2 = 4
illustrative_time_saved_seconds_per_day = 4 * 20 = 80
```

These are user-supplied or trial assumptions to verify, not a product claim. Also compare lifting effort, comfort, actual walkable trips, and whether anyone must move. Do not choose unsafe reachable storage to save time.

Maintenance and limits

Keep numerical heuristics labelled whenever repeated in a proposal. If a source changes, update the affected operational claim as well as the optional catalog. Record exact manual editions and project evidence separately from this generic research.

No exact project product manuals, structural calculations, full accessibility standard, local fire/escape provisions, quantitative acoustic assessment, or photometric design were reviewed to produce this generic skill. Obtain them when the project requires them. A checklist review does not replace installation approval.

living-room-planning: Full Source Catalog

Source file: `.github/skills/living-room-planning/references/source-catalog.md`

SHA-256: `7be2ead08c5f6b7920902af8a22bb7067445d76a5c51f27aabc553b0c50272f0`

Living Room Planning Source Catalog

Optional research/provenance file. Do not load for routine planning or solely to expand a source ID. Load when the user asks for citations, a claim needs verification, sources conflict, or the skill is being maintained.

Research reviewed: 2026-10-09. Scope: generic domestic living-room planning with Germany/EU defaults. Summaries are original planning notes, not reproduced articles. The operational rules and qualifications are in the planning reference; the workflow is in the skill.

S1 - IKEA Germany: activity zones and reset storage

- Source: So gestaltest du drei Arbeitsbereiche in einem Raum.
- Type: retailer guidance; German market.
- Retrieved support: separate work/relaxation areas, visual screening, movable storage, keeping work items together for easy reset.
- Applied: activity and simultaneous-use brief, nearby storage, explicit setup/reset actions.
- Limits: product inspiration, not dimensional or acoustic compliance evidence. Do not generalize the sofa/laptop examples into suitable full-time workstation ergonomics. Screens do not establish speech privacy or sound isolation.

S2 - IKEA Germany: living-room lighting

- Source: Bestmoegliche Nutzung deiner Wohnzimmerbeleuchtung.
- Type: retailer guidance; German market.
- Retrieved support: combine general, task, and mood lighting; support concentrated tasks with appropriate surrounding light; cabinet lighting and adjustable controls.
- Applied: activity-specific lighting, contrast/glare review, multiple controllable sources.
- Limits: does not establish mandatory domestic illuminance, colour temperature, or a required number of lamps. Battery-powered lighting still needs energy and charging; the page's casual wording is not an electrical claim.

S3 - IKEA Germany: TV furniture

- Source: So waehlst du eine TV-Bank, die bestens in dein Zuhause passt.
- Type: retailer guidance; German market.
- Retrieved support: screen centre at seated eye height; open/closed storage choices; cable openings; suggests cabinet wider than screen.
- Applied: seated-eye planning, storage configuration, cable access.
- Limits: furniture wider than the screen does not establish stability. Check feet, depth, loads, anchors, and exact manuals independently. Product range dimensions/prices are not universal design rules and are not embedded in the skill.

S4 - Sony: wall-mounted TV height

- Source: What is the recommended height when mounting the TV on a wall?.
- Type: manufacturer general support; US site. Retrieved page displayed article ID 00026358 and modification date 2019-08-19 despite the requested URL.

- Retrieved support: height depends on preference and seating; generally centre the screen at viewers' eye level; model-specific off-axis image characteristics need checking.
- Applied: actual seated eye measurements, posture trials, per-seat sightline checks.
- Limits: not a fixed floor-height rule, comfort guarantee, bracket specification, or installation manual. Panel viewing-angle specifications differ from the visual angle a screen occupies.

S5 - Sony: viewing distance qualifications

- Source: Recommended viewing distance for watching TV.
- Type: manufacturer general support; US site. Retrieved page displayed article ID 00008602 and modification date 2025-01-17.
- Retrieved support: distance depends on screen size/resolution, content, surroundings, and the viewer; adapt reference distances to preference and room layout.
- Applied: avoid a universal screen-diagonal multiplier; trial the intended screen and content.
- Limits: the fetched text did not expose the numeric content inside the 4K/HD/SD sections. No numerical distance recommendation is attributed to this source. The reference's screen-angle example is elementary geometry, not a Sony recommendation.

S6 - nullbarriere: living/sleeping-room accessibility

- Source: DIN 18040-2 Wohnraum, Schlafraum.
- Context: DIN 18040-2 index, identifying edition 2011-09 and distinguishing ordinary barrier-free use from unrestricted wheelchair use (R).
- Type: German specialist secondary standard summary, not the normative text.
- Retrieved support: turning/movement areas of 1200 x 1200 mm and 1500 x 1500 mm for wheelchair use; depths along/in front of furniture of 900 mm and 1500 mm respectively; flexibility and functional access matter.
- Applied: qualified accessibility prompts and provisional checks; do not import ordinary coffee-table gaps into accessible approaches.
- Limits: current edition, legal adoption by federal state, contractual applicability, exact measurement provisions, and exceptions need verification from authoritative project sources. Full standard text was not reviewed. Bedroom examples and unrelated dimensions are not generalized into living-room rules.

S7 - Verbraucherzentrale: keep radiator heat flow clear

- Source: Heizung optimieren und Heizkosten sparen.
- Type: German consumer advice; retrieved article dated 2026-10-08.
- Retrieved support: long curtains, furniture, and covers in front of radiators can trap heat, impair room warming, and distort thermostat sensing; keep radiators unobstructed.
- Applied: furniture/shading layout must preserve heat flow and usable controls.
- Limits: no universal clearance is inferred. Heating-system modification, fireplace clearances, and cold-wall moisture assessment require their own evidence. Unrelated legislation and numerical savings on this page are not used.

S8 - US CPSC: furniture and TV tip-over prevention

- Source: Anchor It!.
- Type: US government consumer-safety campaign.
- Retrieved support: anchor susceptible furniture and televisions using suitable anchoring equipment; follow installation instructions.
- Applied: include anchoring in layout/installation planning, account for wall construction, and do not assume a low or visually stable unit needs no assessment.

- Limits: US regulations and typical wall construction are not German/EU requirements. Exact product manuals, appropriate substrate/fixings, and professional review control project installation. This skill does not reproduce a generic DIY anchoring procedure.

S9 - US CPSC: window-covering cords

- Source: Window Covering Cords (retrieved canonical destination: Go Cordless).
- Type: US government consumer-safety guidance.
- Retrieved support: cordless coverings are the safest choice around young children; keep furniture that enables access away from windows/cords; mitigation of existing cords does not eliminate the hazard.
- Applied: check climbing routes and reachable cords when positioning sofas, shelves, and play areas; prefer cordless coverings.
- Limits: not a claim about current German/EU product law or a complete child-safety audit. No US retrofit dimensions or regulatory requirements are transposed into the skill.

Research gaps and update policy

- The initially attempted living-room accessibility URL was invalid; S6 records the valid page found through the index, not the failed address.
- An attempted licht.de living-room page yielded no meaningful extract and is not used as evidence.
- The ordinary route, coffee-table, and conversation spacing ranges are explicitly labelled skill heuristics, not research-backed minima.
- Acoustic cautions and delivery/open-state checks are planning synthesis, not validated acoustic performance or model-specific installation data.
- No exact project manuals, current full DIN standard, local fire/escape rules, engineering calculations, or quantitative lighting/acoustic study were inspected.
- For updates, recheck the relevant article and its scope, then update the operational claim and its qualifications. Never replace a missing source with an unqualified mandatory-sounding number.